

claude-windows / c6028e72-ebfd-4218-89a8-c4d7dd23be96

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c6028e72-ebfd-4218-89a8-c4d7dd23be96\subagents\workflows\wf_f9aa1d1c-c83\agent-a7fc5761315556a70.jsonl`
- SHA-256: `938c1cd96de76b3378ee116f41561e17733f6305f66148c74e06d0a91669078b`
- Source modified: `2026-06-09T11:53:38+00:00`
- Imported at: `2026-07-05T16:48:23+00:00`
- project: `wf_f9aa1d1c-c83`
- session_id: `c6028e72-ebfd-4218-89a8-c4d7dd23be96`
```

Transcript

1. user (2026-06-09T11:39:52.111Z)

Adversarially VERIFY the load-bearing claims in these findings, focus: "the VLM-can-reach-high-P/R-temporal-localization claim, the single-model-multisport claim, and the conversational-QA-via-event-store feasibility ? are these supported by real systems/benchmarks or aspirational?". Use WebSearch+WebFetch against primary sources (HF model cards/LICENSE files, official docs, papers/benchmarks). Default to uncertain/outdated if unconfirmed. Flag anything that would change a recommendation.

FINDINGS:

```
[
  {
    "dimension": "BASE MODELS for a single, multisport, conversational video-QA model that also does precise temporal localization (rally/highlight start-end)",
    "summary": "A commercially-clean base must satisfy four things at once: (1) Apache-2.0/MIT/BSD weights with NO viral/NC/MAU restriction; (2) native video understanding of long clips (full match, not a 60s window); (3) emits TIMESTAMPS for moment localization (the rally/highlight start-end requirement); (4) holds multi-turn conversation about an uploaded video (the north-star QA loop). Verified June 2026: Qwen3-VL is the only candidate that hits all four cleanly. It ships Apache-2.0 UNIFORMLY across every size (2B/4B/8B/32B/30B-A3B/235B-A22B ? no per-size license fork, unlike older Qwen generations), with a purpose-built \"Text-Timestamp Alignment\" mechanism for second-level timestamp-grounded event localization, hours-long video via 256K(?1M) context, and native multi-turn multimodal chat. Gemma 4 (released April 2026) is a genuine standard Apache-2.0 release (its Prohibited-Use Policy is non-binding guidance, not a weights restriction ? no MAU cap, no commercial bar), and adds native audio (useful for racquet-sport hit detection) ? but it caps video at ~60 seconds and documents NO temporal-localization/timestamp capability, which directly disqualifies it as the PRIMARY extractor for full-clip start-end localization. InternVL3 is license-clean (MIT project + Apache-2.0 Qwen2.5 LLM) and does multi-round video chat but has no documented native timestamp grounding. Gemma 3 is rejected (viral custom license per the existing guardrail); Gemma 3n is responsible-commercial-license, not clean Apache-2.0, and edge-focused. Llama 4 / DeepSeek-V4 surface in 2026 roundups but Llama carries a custom community license with an MAU cap (not clean) and neither documents racquet-grade temporal grounding. Recommendation: Qwen3-VL as the base family; prototype the harness on the 4B/8B Instruct sizes (fit the 8GB 2070 Super for inference / small-LoRA; rent cloud GPU for full fine-tunes and the larger sizes), keeping the option to scale to 30B-A3B/32B for the quality-first push and later distil down for the deferred on-device goal ? all within one Apache-2.0 family so no re-architecting.",
    "options": [
      {
        "name": "Qwen3-VL (Alibaba) ? Instruct sizes 2B/4B/8B/32B + MoE 30B-A3B/235B-A22B",
        "what": "Multimodal (text+image+video) VLM series, Nov 2025, with a dedicated 'Text-Timestamp Alignment' mechanism (beyond T-RoPE) for precise, second-level timestamp-grounded event localization; 256K context expandable to 1M for hours-long video; native multi-turn multimodal conversation; Thinking variants for reasoning. Active community fine-tuning + video-grounding ecosystem (e.g. Qwen3-VL-Video-Grounding repo/Space).",
        "license": "Apache-2.0 (verified uniform across ALL sizes incl. the 235B ? no per-size
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license fork)",
  "pros": [
    "Only candidate that natively emits timestamps for moment localization ? directly matches the rally/highlight start-end requirement (Text-Timestamp Alignment, second-level indexing)",
    "Apache-2.0 across EVERY size: clean for private proprietary commercial use; no MAU cap, no NC clause; same license at 2B and 235B means no re-architecting when scaling quality up or distilling down for the deferred on-device goal",
    "Handles hours-long video (256K?1M context) ? a full match clip, not a 60s window ? so it can both localize and hold the contextual multi-turn QA the north-star feature needs",
    "Wide size ladder lets the 8GB 2070 Super run 4B/8B for inference and small LoRA locally, while cloud-rented GPUs do full fine-tunes / larger sizes ? fits the solo-founder + per-job cloud budget",
    "Sport-agnostic by construction (general VLM): one model can be fine-tuned across all net-separated racquet sports, supporting the single sport-agnostic model goal without per-sport bases",
    "Existing open fine-tuning + video-grounding tooling de-risks the training-harness build (the immediate deliverable)"
  ],
  "cons": [
    "Base weights' training-data provenance is Alibaba's and not auditable; the Apache-2.0 license governs YOUR use, but if the owner extends the 'never train on paid-LLM' purity test to the base's own pretraining, that cannot be fully verified (true of essentially every open VLM ? treat as base-license-governed, not a YOUR-training-data issue)",
    "No native audio input (unlike Gemma 4) ? racquet-hit audio cues, if wanted, need a separate audio branch or fusion",
    "Largest MoE sizes (235B) are impractical on the local box and pricey to fine-tune in the cloud; the realistic working set is 4B?32B",
    "Quality on racquet-sport temporal localization is unproven for this corpus ? the de-risking evidence (LOVO ~0.583) is from a separate per-frame prototype, not from this base; needs its own fine-tune + eval"
  ],
  "fit": "strong"
},
{
  "name": "Gemma 4 (Google) ? E2B/E4B/12B/26B-A4B/31B",
  "what": "Google's April 2026 open-weight family; unified multimodal (text+image+video) with NATIVE AUDIO on E2B/E4B/12B; multi-turn chat + system role; 128K?256K context. Released specifically under standard Apache-2.0 (a deliberate break from Gemma 3's custom license).",
  "license": "Apache-2.0 (genuine standard Apache-2.0; the separate Prohibited-Use/Intended-Use docs are non-binding policy guidance ? no MAU cap, no commercial-use bar)",
  "pros": [
    "Truly clean standard Apache-2.0 ? resolves the Gemma-3 viral-license rejection; commercially usable in private proprietary software with no MAU cap",
    "Native AUDIO input (E2B/E4B/12B) ? directly useful for racquet-sport shot/hit detection, a cue the badminton work already explored",
    "Strong size ladder incl. genuinely edge-friendly E2B/E4B for the eventual deferred on-device compression goal",
    "Backed by Google with first-class tooling, quantization, and long-term support"
  ],
  "cons": [
    "DISQUALIFYING for the primary extractor: documents only ~60-second video input and NO temporal-localization/timestamp capability ? cannot emit rally/highlight start-end over a full match clip out of the box",
    "60s window also constrains the contextual multi-turn QA over a whole uploaded clip (the north-star) without heavy external chunking/bookkeeping",
    "Temporal grounding would have to be taught from scratch via fine-tuning against a base not designed to output timestamps ? higher risk/effort than Qwen3-VL's native support"
  ],
  "fit": "weak"
},
{
  "name": "InternVL3 (OpenGVLab) ? e.g. InternVL3-8B",
  "what": "Open multimodal series (April 2025) with strong video-understanding benchmarks and documented multi-round video conversation; built on a Qwen2.5 LLM backbone. Successor lines

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(InternVL3.5) add temporal-grounding gains via add-on methods.",
  "license": "MIT (project) + Apache-2.0 (Qwen2.5 LLM component) ? both clean for
proprietary commercial use",
  "pros": [
    "License-clean for commercial proprietary use (MIT + Apache-2.0)",
    "Strong video-understanding benchmarks and native multi-round video chat ? fits the
conversational QA loop",
    "Good size range; well-supported open ecosystem"
  ],
  "cons": [
    "No NATIVE timestamp/temporal-localization output documented in the base 3.x card ?
timestamp grounding relies on add-on methods (e.g. GroundVTS) or fine-tuning, weaker out-of-box
than Qwen3-VL for the start-end requirement",
    "Dual-license means tracking obligations of both the MIT project and the Apache-2.0
backbone (minor, but a compliance step)",
    "Effectively trails Qwen3-VL on the single most load-bearing requirement (native
timestamp emission)"
  ],
  "fit": "medium"
},
{
  "name": "Gemma 3 / Gemma 3n (Google, earlier)",
  "what": "Prior Gemma generation. Gemma 3 = multimodal but under the custom Gemma
license; Gemma 3n = edge/mobile multimodal (incl. audio) under a 'responsible commercial use'
license.",
  "license": "Gemma custom license (Gemma 3 ? viral/custom terms) / Gemma 'responsible
commercial use' (3n) ? NOT clean Apache-2.0/MIT/BSD",
  "pros": [
    "Gemma 3n is genuinely edge/on-device-oriented with audio ? relevant to the LATER
compression goal",
    "Mature tooling"
  ],
  "cons": [
    "FAILS the hard commercial-license guardrail: Gemma 3's custom/viral terms are
explicitly rejected by the owner; Gemma 3n's 'responsible commercial use' license is not a clean
Apache-2.0/MIT/BSD",
    "Superseded by Gemma 4's clean Apache-2.0 release anyway ? no reason to adopt the
restricted predecessor",
    "No advantage over Qwen3-VL on temporal localization"
  ],
  "fit": "weak"
},
{
  "name": "Llama 4 / DeepSeek-V4 (2026 general-LLM roundup mentions)",
  "what": "Large 2026 models surfaced in open-LLM roundups; Llama 4 Scout cited for very
long video context; DeepSeek-V4 variants cited as MIT with 1M context.",
  "license": "Llama = custom Llama Community License with an MAU cap (NOT clean);
DeepSeek-V4 = MIT (clean) per roundups",
  "pros": [
    "DeepSeek-V4 MIT licensing is commercially clean with very long context",
    "Llama 4 cited for long-video context windows"
  ],
  "cons": [
    "Llama Community License carries an MAU-cap / acceptable-use restriction ? fails the
'reject >X-MAU caps if binding' and clean-license guardrail; not adoptable as the base",
    "Neither documents racquet-grade NATIVE timestamp/temporal-localization for video
moment localization ? primary requirement unmet",
    "DeepSeek-V4's video-VLM + timestamp-grounding maturity is unverified vs Qwen3-VL's
purpose-built support",
    "Roundup-level sourcing only; not validated on the load-bearing temporal-grounding
requirement"
  ],
  "fit": "weak"
}
],

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"recommendation": "Adopt Qwen3-VL as the base-model family. It is the only verified June-2026 option that simultaneously satisfies all four load-bearing requirements: clean Apache-2.0 weights (uniform across every size ? no per-size license fork, no MAU cap, no NC clause), native long-video understanding (256K?1M context = full match clips, not a 60s window), purpose-built timestamp emission for moment localization (its 'Text-Timestamp Alignment' giving second-level event localization ? a direct match for rally/highlight start-end), and native multi-turn conversational video QA (the north-star loop). Build the training harness on the Instruct line: prototype on 4B/8B (runs on the 8GB 2070 Super for inference and small LoRA; full fine-tunes go to rented cloud GPU), reserve 30B-A3B/32B for the quality-first push, and keep the smaller sizes as the distillation target for the DEFERRED on-device goal ? all inside one Apache-2.0 family so scaling up or compressing down needs no re-architecting. Keep Gemma 4 explicitly on the bench for ONE reason: its native audio input (E2B/E4B/12B) is the cleanest path to racquet-hit audio cues if a fusion signal is later wanted ? but it cannot be the primary extractor (60s video cap, no timestamp output). Treat InternVL3 as the license-clean fallback if a Qwen-specific blocker emerges. Two scope notes for the harness, independent of base choice: (1) the conversational-QA north-star is best served by keeping EXTRACTION (the fine-tuned Qwen3-VL emitting timestamped rally/event spans) separate from the QA/RETRIEVAL layer (durable per-video bookkeeping + query routing over those spans) ? the model emits structured spans, the framework stores and answers over them; design that bookkeeping schema now (as the owner asks) but PUNT shipping conversational QA until rally/highlight quality is solid, since folding QA into the same fine-tune competes for the quality budget. (2) Base choice does NOT resolve the WASB checkpoint provenance blocker ? that detector sits below the VLM and stays an independent Gen-0 prerequisite.",

"confidence": "high",

"sources": [

"https://github.com/QwenLM/Qwen3-VL",
"https://huggingface.co/Qwen/Qwen3-VL-8B-Instruct",
"https://huggingface.co/Qwen/Qwen3-VL-2B-Instruct",
"https://huggingface.co/Qwen/Qwen3-VL-235B-A22B-Instruct",
"https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct",
"https://arxiv.org/abs/2511.21631",
"https://deepwiki.com/QwenLM/Qwen3-VL/5.4-video-understanding",
"https://github.com/PRITHIVSAKTHIUR/Qwen3-VL-Video-Grounding",
"https://ai.google.dev/gemma/docs/core",
"https://ai.google.dev/gemma/docs/core/model_card_4",
"https://ai.google.dev/gemma/apache_2",
"https://ai.google.dev/gemma/prohibited_use_policy",
"https://ai.google.dev/gemma/docs/gemma-3n",
"https://en.wikipedia.org/wiki/Gemma_(language_model)",
"https://huggingface.co/OpenGVLab/InternVL3-8B",
"https://arxiv.org/pdf/2504.10479",
"https://llm-stats.com/models/qwen3-vl-8b-instruct"

],

"caveats": [

"Per-size Apache-2.0 uniformity for Qwen3-VL was verified on the 2B, 4B(referenced), 8B, and 235B HF cards plus the GitHub repo footer ? all read apache-2.0; I did not individually open every single size/edition (e.g. 32B-Thinking, 30B-A3B FP8), but the consistency across the smallest and largest plus the repo-level license makes a per-size fork very unlikely. Re-confirm the exact size you ship before release.",

"Gemma 4's Prohibited-Use Policy was assessed as non-binding policy guidance (no MAU cap, no commercial bar) from the policy page itself; if legal certainty is required, have counsel confirm whether Google treats it as a binding condition on the Apache-2.0 grant. This does not change the recommendation (Gemma 4 is disqualified on the 60s/no-timestamp technical limits regardless).",

"The Gemma 4 ~60-second video cap and 'no temporal localization' come from the official model card/overview as of this search; Google could extend these ? recheck if Gemma 4 becomes a serious contender. Not load-bearing for the recommendation.",

"Base-model PRETRAINING data provenance (for ALL candidates, incl. Qwen3-VL) is not publicly auditable. The owner's 'never train on paid-LLM outputs' guardrail is satisfied at the level of YOUR fine-tuning data; whether a vendor's base pretraining touched paid-LLM outputs is unknowable and is governed by the base's license, not your training pipeline. Flag for owner's risk acceptance.",

"Qwen3-VL's quality on THIS racquet-sport temporal-localization task is unproven ? the cited de-risking evidence (LOVO ~0.583, AUC 0.83?0.92) is from a separate per-frame prototype, not this base. The recommendation is on FIT/license/capability, not a benchmarked accuracy

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claim; the harness must run its own fine-tune + cross-video eval.",
    "2026 'best open LLM' roundups (Llama 4, DeepSeek-V4) are secondary/blog sources and were
not individually license-verified at the model-card level; they are included only to show they
were considered and ruled out on the temporal-grounding and (for Llama) MAU-cap grounds, not as
adoptable candidates."
]
},
{
    "dimension": "Fine-tuning frameworks for video VLMS (SFT + RFT/GRPO for temporal grounding +
conversational video-QA)",
    "summary": "FIRST, THE ORTHOGONALITY THE OWNER ASKED ME TO CLARIFY: a *framework* (ms-swift,
LLaMA-Factory, axolotl, unsloth, TRL) and a *base model* (Qwen3-VL, Qwen2.5-VL, Qwen3-Omni,
Gemma-4) are independent choices. The framework is the \"forge\" that runs SFT/GRPO on a GPU and
emits LoRA/full weights; the base is the metal you forge. You pick ONE framework and feed it ANY
Apache/MIT base, and a different base later (Qwen->Gemma) is a config swap, not a re-
platforming. So this dimension does NOT pick the model; it picks the trainer. ALL FIVE leading
frameworks are permissively licensed (Apache-2.0; unsloth's only AGPL piece is the optional
Studio UI, trivially avoided), so the hard commercial guardrail does not eliminate any of them
at the framework layer; license cleanliness is decided by the BASE WEIGHTS + DATA, not the
trainer. The differentiator is capability on the four hard requirements: (1) NATIVE video
training, (2) NATIVE audio training, (3) timestamp/temporal-grounding outputs, (4) RFT/GRPO with
a CUSTOM reward ON VIDEO inputs, runnable on a rented A100/L40S. On those four jointly, exactly
one framework clears all four in a single stack: ms-swift. It is the only framework that
advertises and documents native video+audio training AND multimodal GRPO with a custom external-
reward plugin executed over video rollouts (via vLLM), plus the fps/num_frames video-sampling
controls you need to train timestamped grounding. The others each fail at least one hard
requirement: LLaMA-Factory does native video+audio SFT but DELEGATES GRPO to a separate repo
(EasyR1) that is image-only; unsloth's much-praised Vision-RL/GRPO is image-only (its value is
the 8GB-box SFT, which matters LATER for compression, not now); TRL's GRPO-VLM is image-only and
you would hand-build the video reward loop; axolotl explicitly labels video \"not well tested.\"
NOTE ON TEMPORAL GROUNDING: emitting timestamps is primarily a BASE-MODEL + dataset-format
property (Qwen2.5/3-VL align MRoPE position IDs to wall-clock time; you train on a JSON messages
format with second/HMS timestamps), so the framework's job is just to expose video-frame
controls and not corrupt the time encoding, which ms-swift does and is the most-documented for
it. RECOMMENDATION: adopt ms-swift as the single training/RFT forge now; keep unsloth in the
back pocket as the 8GB-local SFT/compression tool for the deferred on-device goal. This aligns
with, and now hardens, the lean already recorded in docs/PLATFORM_ARCHITECTURE.md (which favored
ms-swift but predates the RFT-custom-reward and audio requirements that make the gap
decisive).",
    "options": [
        {
            "name": "ms-swift (ModelScope SWIFT)",
            "what": "Self-managed CLI/Python fine-tuning + RL + deploy framework, 600+ LLMs / 300+
MLLMs. The ONLY framework that natively trains video AND audio (Qwen3-VL, Qwen3-Omni, Gemma-4,
InternVL3.5) and runs multimodal GRPO with a custom external-reward plugin over video rollouts
(vLLM-accelerated). Exposes fps/num_frames video sampling; JSON messages dataset format carries
timestamps for temporal grounding. Has an eval backend (EvalScope) and args.json
reproducibility, fitting the agent-runnable report-card harness goal.",
            "license": "Apache-2.0",
            "pros": [
                "ONLY framework meeting all four hard requirements in one stack: native video + native
audio + timestamp/grounding controls + GRPO/RFT with a CUSTOM reward on VIDEO inputs (documented
Qwen2.5-VL GRPO example uses --external_plugins + --reward_funcs + --use_vllm)",
                "Apache-2.0, clean for private/proprietary/commercial use at the framework layer; no
MAU cap or NC clause",
                "Supports the exact bases on your approved list (Qwen3-VL, Qwen2.5-VL, Qwen3-Omni for
audio, Gemma-4) so the framework decision does not constrain the later base choice",
                "Any model size incl. cheap-to-serve 7B/4B; runs on rented A100/L40S (your
Modal/RunPod plan); multimodal GRPO examples are 6-8 GPU but SFT/LoRA fits a single 24-48GB
card",
                "Custom reward plugin is the seam to reuse your existing metrics.match_intervals
temporal-IoU scorer as the RFT reward (the study Gen-2 plan): single scorer spine, no metric
drift",
                "Reproducibility (args.json, full_determinism) + EvalScope eval backend map onto the
nanochat-style one-command report-card harness you want",
            ]
        }
    ]
}

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"Active, large project (AAAI 2025; v4.0 Mar-2026), broadest model coverage of any
option"
],
"cons": [
  "Heaviest/most-configurable surface area, many flags; steeper first-run than unsloth
notebooks",
  "Multimodal GRPO is VRAM-hungry (multiple video rollouts/sample); real video-token
VRAM on 1080p/60fps clips is UNVERIFIED (same open risk flagged in the training study); re-
benchmark before budgeting",
  "Audio-modality (Omni) training is less battle-tested than its video/image path;
verify the Qwen3-Omni recipe before relying on audio cues",
  "ModelScope-centric defaults (some model pulls assume ModelScope hub); point it at HF
mirrors for clean provenance"
],
"fit": "strong"
},
{
  "name": "LLaMA-Factory (+ EasyR1 for RL)",
  "what": "Popular, well-documented self-managed SFT/DPO framework for 100+ LLMs/VLMs with
native video (LLaVA-NeXT-Video) and audio (Qwen2-Audio, Qwen2.5-Omni) SFT. GRPO is NOT in-core;
it is delegated to the sibling repo EasyR1 (a veRL fork), which is image/text-only.",
  "license": "Apache-2.0 (both LLaMA-Factory and EasyR1)",
  "pros": [
    "Apache-2.0, very approachable, best-documented onboarding; native video + audio SFT
in-core",
    "Strong if your near-term need is SFT-only cold-start (Gen-2 SFT cold-start step)
before any RL",
    "Large community, many tutorials, lowest ramp for a solo founder doing supervised
runs"
  ],
  "cons": [
    "DECISIVE GAP: GRPO/RFT is delegated to EasyR1, which is IMAGE-ONLY (no video, no
audio, no confirmed custom-reward-on-video path). So your RFT-with-temporal-IoU-reward-on-video
requirement is NOT met in one stack; you would run two frameworks and EasyR1 still cannot take
video",
    "Splitting SFT (LLaMA-Factory) and RL (EasyR1) doubles the harness surface and breaks
the single-command report-card goal",
    "Video SFT works but the framework is not the strongest-documented for fine-grained
timestamp grounding vs ms-swift Qwen-VL best-practices"
  ],
  "fit": "medium"
},
{
  "name": "Unsloth",
  "what": "Single-GPU-optimized framework (70-90% less VRAM, 1.5-2x faster) with Qwen3-VL
SFT and a celebrated Vision-RL (GRPO/GSP0) path. Trains Qwen3-VL-8B GRPO on a free Colab T4. The
RL path is IMAGE-input only.",
  "license": "Apache-2.0 core (optional Unsloth Studio UI is AGPL-3.0, avoidable; do not
ship the Studio component)",
  "pros": [
    "Best-in-class for the 8GB RTX 2070 Super: can SFT/LoRA a 7B/8B VLM locally where
others OOM, directly serving the DEFERRED on-device/compression goal",
    "Apache-2.0 core; vision GRPO/GSP0 supported (image), 256K context",
    "Cheapest first runs / fastest local iteration; great for local SFT cold-start and
later quantization/compression"
  ],
  "cons": [
    "Vision-RL is IMAGE-ONLY, does not satisfy the RFT-with-custom-reward-ON-VIDEO
requirement central to the quality goal",
    "Single-GPU focus; not the multi-GPU rented-A100/L40S RFT engine for video",
    "Tightly coupled to its own patched kernels/model classes, less of a general swap-any-
base + any-reward forge than ms-swift",
    "Keep the AGPL Studio UI out of the proprietary product to stay license-clean"
  ],
  "fit": "medium"
}

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    },
    {
      "name": "HuggingFace TRL (+ transformers)",
      "what": "The foundational RL/post-training library (GRPO, DPO, PPO, RL00) that most
other frameworks build on. Multiple weighted custom reward functions are first-class. Its
documented GRPO-VLM path is image+text.",
      "license": "Apache-2.0",
      "pros": [
        "Apache-2.0, maximally portable, the de-facto substrate; owning the recipe here is the
ultimate anti-lock-in hedge",
        "Cleanest, most flexible CUSTOM reward API (list of reward fns, reward_weights), ideal
abstraction for your temporal-IoU reward",
        "Tight transformers integration; any HF Apache/MIT base loads directly with clean
provenance"
      ],
      "cons": [
        "GRPO-VLM is documented/tested for IMAGE inputs; video rollout plumbing (frame
sampling, video reward batching) is DIY, you become the framework author",
        "No batteries-included native video/audio dataset pipeline or fps/num_frames
ergonomics, slowest path to a working video RFT loop",
        "Lower-level: more glue code for a solo founder vs ms-swift ready video-GRPO examples"
      ],
      "fit": "medium"
    },
    {
      "name": "Axolotl",
      "what": "Production-oriented YAML-config SFT/DPO/GRPO framework with a BETA multimodal
path (Qwen2-VL, Pixtral, SmolVLM2 video, Voxtral/Gemma-4 audio).",
      "license": "Apache-2.0",
      "pros": [
        "Apache-2.0; clean reproducible YAML configs, good production ergonomics for SFT",
        "Has audio (Voxtral, Gemma-4) and some video models; LoRA/QLoRA efficient"
      ],
      "cons": [
        "DECISIVE GAP: video support is explicitly documented as not well tested at the
moment, the wrong bet for a VIDEO-FIRST very-high-quality goal",
        "Multimodal is framework-level BETA; custom-reward-on-video GRPO not demonstrated",
        "Strengths (text SFT ergonomics) are not where this project hard requirements live"
      ],
      "fit": "weak"
    }
  ],
  "recommendation": "Adopt ms-swift (Apache-2.0) as the single SFT+RFT forge now. It is the
only framework that meets all four hard requirements jointly: native video, native audio
(Qwen3-Omni/Gemma-4), timestamp/grounding via fps/num_frames + the JSON messages format, and
GRPO/RFT with a CUSTOM external-reward plugin executed over VIDEO rollouts (vLLM); and it runs
on the rented A100/L40S you already plan to use (Modal/RunPod). Its custom-reward plugin is the
exact seam to wire your existing metrics.match_intervals temporal-IoU scorer in as the RFT
reward (the Gen-2 plan in OWNED_MODEL_TRAINING_STUDY.md), keeping one scorer spine across
heuristic-labeling, eval, and RL. The competitors each fail a hard requirement: LLaMA-Factory
does native video+audio SFT but offloads GRPO to EasyR1, which is image-only (so the video-RFT
goal needs a two-framework Frankenstein that still cannot take video); unsloth's Vision-RL and
TRL's GRPO-VLM are both image-only; axolotl's video path is self-described as not well tested.
CLARIFY FOR THE OWNER (the orthogonality you asked me to state): this decision is independent of
the base-model choice; ms-swift trains Qwen3-VL, Qwen2.5-VL, Qwen3-Omni (audio), and Gemma-4
alike, so which-framework and which-base are decoupled and the base can change later without re-
platforming. Because every leading framework is permissive, the commercial-license guardrail is
NOT decided here; it is decided by the base WEIGHTS provenance and the training DATA (your WASB-
checkpoint C3 blocker and the no-paid-LLM-labels rule live at the data/weights layer, not the
trainer). Keep unsloth as the secondary, 8GB-local tool for the DEFERRED on-device/compression
goal (it is the best fit for the RTX 2070 Super), and keep TRL in view as the portability hedge
since ms-swift, unsloth, and EasyR1 all sit on TRL/veRL-style primitives anyway; you own the
data + recipe + exported weights, so the forge stays swappable. ON CONVERSATIONAL VIDEO-QA /
BOOKKEEPING: the framework choice does not force the embark-now vs punt decision; the per-video
state (rally inventory, shot counts, fault tags, derived clips) is an APPLICATION/RETRIEVAL

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layer that NO trainer holds; ms-swift just lets you later add a QA-style SFT task over the same corpus when quality is solid. So pick the forge now, build the bookkeeping store from the start as the brief requires, and keep conversational-QA as an SFT task you can add to ms-swift later rather than a framework prerequisite.",

```
"confidence": "high",
"sources": [
  "https://github.com/modelscope/ms-swift",
  "https://swift.readthedocs.io/en/latest/BestPractices/GRPO-Multi-Modal-Training.html",
  "https://qwen.readthedocs.io/en/latest/training/ms_swift.html",
  "https://github.com/modelscope/ms-swift/issues/7326",
  "https://github.com/hiyouga/LLaMA-Factory/blob/main/README.md",
  "https://github.com/hiyouga/EasyR1",
  "https://docs.axolotl.ai/docs/multimodal.html",
  "https://www.spheron.network/blog/axolotl-vs-unsloth-vs-torch-tune/",
  "https://unsloth.ai/docs/get-started/reinforcement-learning-rl-guide/vision-reinforcement-learning-vlm-rl",
  "https://unsloth.ai/docs/models/qwen3-how-to-run-and-fine-tune/qwen3-vl-how-to-run-and-fine-tune",
  "https://github.com/unslothai/unsloth",
  "https://github.com/huggingface/trl/blob/main/docs/source/grpo_trainer.md",
  "https://huggingface.co/learn/cookbook/fine_tuning_vlm_grpo_trl",
  "https://pyimagesearch.com/2025/06/16/video-understanding-and-grounding-with-qwen-2-5/",
  "https://arxiv.org/pdf/2505.20715",
  "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/PLATFORM_ARCHITECTURE.md",
  "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md"
],
"caveats": [
  "VRAM/token cost of multimodal GRPO on real 1080p/60fps clips is UNVERIFIED for every framework (rollouts multiply video-token cost); the documented ms-swift multimodal GRPO examples use 6-8 GPUs; re-benchmark on a rented A100/L40S before budgeting. This is the same open risk already flagged in OWNED_MODEL_TRAINING_STUDY.md, not new.",
  "Framework license is clean across the board, but that does NOT clear the COMMERCIAL guardrail; license cleanliness is decided by the BASE WEIGHTS provenance (e.g. the unresolved WASB checkpoint C3 blocker) and the TRAINING DATA (no paid-LLM labels; minors excluded). Do not let framework-is-Apache-2.0 imply the pipeline is clean.",
  "ms-swift audio (Qwen3-Omni / Gemma-4) training path is less battle-tested than its image/video path; verify the Omni recipe end-to-end before committing to audio as a training cue. Audio was NOT adopted for fusion in prior work (PR #35, audio-E1 findings), so treat audio-in-training as exploratory.",
  "Temporal-grounding/timestamp emission is mostly a BASE-MODEL + dataset-format property (Qwen MROPE time alignment), not a framework feature; choosing ms-swift gives you the controls (fps/num_frames, timestamped messages) but the grounding QUALITY depends on the base and the data format, which must be validated separately.",
  "Web specifics (version numbers, exact model lists, GPU counts in examples) are 2026 snapshots and move fast; re-confirm the live ms-swift docs/version at implementation time. Implementation remains gated on owner sign-off + PLATFORM scaffolding per the existing sequencing directive; this is a framework SELECTION, not a go-build order.",
  "EasyR1 being image-only is the load-bearing reason LLaMA-Factory drops to medium; if EasyR1 (or LLaMA-Factory in-core) adds video GRPO later, re-evaluate; LLaMA-Factory SFT ergonomics are otherwise excellent for the cold-start step."
]
},
{
  "dimension": "Model strategy for high-P/R rally+highlight temporal extraction (multisport, single model): fine-tuned VLM vs specialized TAL head vs both",
  "summary": "The evidence says BOTH, in a deliberate sequence ? and it strongly validates the project's existing 3-generation blueprint (TAS head now ? VLM later) while sharpening the rationale. The core empirical fact: a fine-tuned VLM does NOT reach \"very high\" precision/recall at STRICT temporal boundaries. The strongest 2025-26 RL-post-trained video VLM (Time-R1 on Qwen2.5-VL-7B) reaches only R@0.7=50.1% mIoU=60.9% on Charades-STA *fine-tuned*; the same base Qwen2.5-VL-7B off-the-shelf is R@0.7=24.3%. Specialized TAL heads remain ahead on boundary precision (ActionFormer ~66.8% mAP on the harder multi-instance THUMOS14; Dec-2025 TBT-Former 68.0%). The literature consensus is explicit: VLMs \"capture coarse temporal regions rather than accurate event boundaries\" and \"struggle with absolute fine-grained temporal
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grounding\" ? which is exactly the rally start/end accuracy this product sells. So: the TAL/own-feature head is the right vehicle for the *quality-first* extraction goal NOW (cheap, \$0 local on the 2070, already de-risked to AUC 0.84-0.91 / LOVO 0.615 at n=3), and the VLM is the right vehicle for the *conversational-QA north star* LATER ? but evidence says the QA layer should be RAG/event-index over the head's structured output, NOT a single end-to-end VLM that does both detection and conversation. A single sport-agnostic model is feasible and even beneficial: RacketVision (AAAI'26, badminton+tennis+TT) shows unified multi-sport training BOOSTS generalization +15-19% mAP, at a modest cost to per-sport pinpoint precision ? directly answering the multisport tension. Recommendation: BOTH, sequenced ? own TAL head as the quality-first extractor + teacher/label-amplifier and the honest baseline; VLM as the deferred conversational target fed by the head's event index. Embark on the bookkeeping/event-index framework NOW (it's no-regret and the QA layer needs it regardless); PUNT the end-to-end conversational VLM until extraction quality is solid and the corpus is large.",

```
"options": [
  {
    "name": "Option A ? Specialized TAL head over own feature stack
(ActionFormer/ASFormer/TriDet), per-sport-agnostic, single shared head",
    "what": "A tiny temporal-action-localization/segmentation head (ActionFormer 1:1
rally=instance, or ASFormer/MS-TCN2 for per-frame TAS) trained on a hand-built per-frame feature
stack derived from a ball/shuttle trajectory detector (WASB-class) plus motion/audio/pose cues.
One shared, sport-agnostic head; sport identity is just another input feature or a per-sport
calibration, not a separate model. This is the project's de-risked Gen-0/Gen-1 path.",
    "license": "MIT (ActionFormer, ASFormer, TriDet, MS-TCN2 ? use sj-li/MS-TCN2 not the
Commons-Clause yabufarha repo). Trajectory detector code MIT (WASB) but the distributed
checkpoint provenance is UNRESOLVED (C3 blocker) ? own-weights or a clean alternative needed.",
    "pros": [
      "Highest boundary precision per dollar ? specialized heads still lead VLMs at strict
IoU (ActionFormer ~66.8% mAP THUMOS14 multi-instance; TBT-Former 68.0% Dec-2025). Boundary
accuracy IS the product.",
      "Already de-risked on THIS corpus: learned per-frame prototype AUC 0.84-0.91 held-out,
LOVO mean 0.615 at n=3, overtaking the heuristic and decoupling from the multi-court confound.",
      "Trains in minutes, ~4.5GB VRAM, $0 on the local 2070 Super ? fits the solo-
founder/single-GPU budget; no cloud DPA needed for training.",
      "Tiny (~1M params) ? trivially on-device later, satisfying the deferred compression
goal without re-architecting.",
      "Sport-agnostic single head is validated: RacketVision shows unified multi-sport
training improves generalization +15-19% mAP. The feature stack (trajectory
visibility/velocity/net-crossing) is a racket-sport invariant.",
      "Doubles as the label-amplifier/teacher (confidence-gated pseudo-labels) and the
honest baseline the VLM must beat ? pure upside even if the VLM later wins."
    ],
    "cons": [
      "Cannot do conversation / open-ended QA ? it only emits [start,end,label]; the north-
star QA feature must live in a separate retrieval/LLM layer on top.",
      "Generalization to tennis/TT/padel is asserted-but-unproven on THIS stack: the corpus
and WASB features are badminton-only today; needs per-sport trajectory detectors (clean-license
ones not yet identified for all 5 sports) + per-sport labeled data.",
      "Depends on a clean trajectory detector; WASB checkpoint provenance (C3) is an open
commercial-cleanliness blocker that must be resolved (own-train weights or swap, e.g. MS-
TrackNetV3-class) before commercial ship.",
      "Feature engineering is per-sport-fragile at the edges (net height, ball speed regimes
differ) even if the schema generalizes ? some per-sport calibration is likely."
    ],
    "fit": "strong"
  },
  {
    "name": "Option B ? Fine-tuned single end-to-end VLM (Qwen3-VL-4B/Qwen2.5-VL-7B) doing
detection + segmentation + conversational QA",
    "what": "One Apache-2.0 video VLM fine-tuned to emit rally [start,end] timestamps AND
answer contextual questions, via SFT cold-start then GRPO/RFT with a temporal-IoU-vs-golden
reward. The owner's 'hopefully one conversational model' end-state.",
    "license": "Apache-2.0 bases exist and are clean: Qwen3-VL-4B/8B, Qwen2.5-VL-7B/32B,
Gemma-4 E4B/12B-Unified (keep AMBER until raw HF LICENSE verified). BUT the standard public
grounding-CoT training data (e.g. TVG-R1's 56K) is Gemini-2.5-Pro-generated ? FORBIDDEN here;
you must build your own CoT labels, which is the binding cost.",
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    "pros": [
        "Only path to the conversational-QA north star within a single model ? answers 'show me the longest rally', 'all the service faults', natural-language slicing.",
        "Generalizes across sports/play-types with far less per-sport feature engineering ? semantic priors transfer; one model, many sports, fewer bespoke detectors.",
        "Rapidly improving: 2025-26 RL post-training (Time-R1, MUSEG, TimeRFT) lifted Qwen2.5-VL-7B from R@0.7=24.3% (off-the-shelf) to 50.1% fine-tuned; Qwen3-VL adds explicit text-timestamp alignment for second-level indexing.",
        "Same labeled corpus feeds it via the clip2QA transcoder ? no re-labeling, consistent with the moat design."
    ],
    "cons": [
        "Does NOT hit 'very high' P/R at strict boundaries: best fine-tuned VLM is R@0.7?50% mIoU?61% on Charades-STA ? half of moments still misaligned at IoU 0.7. The literature is explicit VLMs 'capture coarse regions, not accurate boundaries.' This is a direct miss on the quality-first goal.",
        "Forbidden-teacher penalty is real and binding: the proven data recipe (~13K SFT + ~18K RL CoT) leans on paid-LLM-generated CoT labels, which this project bans ? you must originate that supervision yourself, raising the bar substantially.",
        "Cannot train locally: QLoRA of a 7B video VLM needs ~24GB+; requires rented cloud (Modal/RunPod) + signed DPA/no-train for consented footage. GRPO multiplies cost (many rollouts over long visual contexts).",
        "On-device path is heavier (4B quantized on 8GB Turing is unbenchmarked) ? defers, not deletes, the compression problem.",
        "Holding per-video QA STATE (rally inventory, shot counts, fault tags) is NOT something the model itself does reliably ? even the single-model dream still needs external bookkeeping, so it doesn't eliminate the framework requirement."
    ],
    "fit": "weak"
},
{
    "name": "Option C (RECOMMENDED) ? BOTH, sequenced: own TAL head as quality-first extractor + teacher/baseline; VLM deferred as the conversational target fed by a RAG/event-index over the head's output",
    "what": "Phase 1: ship the sport-agnostic TAL head as the high-P/R rally/highlight extractor (the de-risked Gen-0/1). It writes a durable per-video EVENT INDEX (rally inventory, [start,end], shot counts, ending_reason/fault tags, derived-clip refs). Phase 2 conversational QA = an LLM/agent doing RETRIEVAL over that index (AVAS-style event knowledge graph + RAG), composing answers/clips deterministically ? NOT re-deriving events. The head also serves as a confidence-gated pseudo-label teacher and the honest baseline. A fine-tuned VLM is a LATER, optional upgrade for the grounding/QA stage once the corpus is large and quality is solid ? and even then likely as a coarse-proposer refined by the head (the TimeRefine/perception-decoder two-stage pattern), not a wholesale replacement.",
    "license": "All-clean: MIT TAL heads + Apache-2.0 LLM for the QA/retrieval layer (Qwen-class) + own/clean trajectory weights. No paid-LLM in any TRAINING path; paid Gemini stays inference/fallback only. The QA-layer LLM does inference-time reasoning over YOUR structured index (allowed) ? it is not trained on paid-LLM outputs.",
    "pros": [
        "Hits the quality-first goal NOW with the boundary-accurate head, while keeping the conversational north star reachable ? no false choice between them.",
        "RAG-over-event-index is the 2025-26 best practice for long-video contextual QA (AVAS event knowledge graph, MemoryCard): more accurate, cheaper, and auditable than end-to-end VLM QA, and it's deterministic for 'make a clip of all X'.",
        "Satisfies 'bookkeeping from the start': the event index IS the per-video state the QA loop needs ? build it now as a no-regret framework artifact even if QA ships later.",
        "Decouples risk: extraction quality and conversational capability evolve independently; you never block shipping a good extractor on an unproven VLM, and you can A/B a VLM proposer behind the same scorer spine.",
        "Cheapest commercially-clean route: $0 local training for the head; the QA LLM is inference-only over your data; the VLM cloud-training spend is deferred until justified by QA volume and a $/call crossover.",
        "Same one corpus + one scorer (match_intervals / tIoU) feeds head training, TAL eval, pseudo-labeling, AND the eventual VLM reward ? zero metric drift, the moat stays the dataset+harness."
    ],

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    "cons": [
        "Two systems to build and integrate (extractor + retrieval/QA layer) rather than one
model ? more surface area, a clear data contract (the event-index schema) is required up
front.",
        "The conversational answers are only as good as the head's event tags ? richer
semantics (e.g. 'service fault') need explicit event types the head/labels must support, adding
label taxonomy work.",
        "Defers the 'single conversational model' aesthetic the owner hopes for; if a single-
model VLM later proves it can do both at high P/R, some of the retrieval scaffolding is
superseded (but the index/corpus is reused, so low regret).",
        "Still inherits Option A's open blockers: WASB C3 provenance and multisport
feature/detector generalization must be resolved."
    ],
    "fit": "strong"
}
],
"recommendation": "Adopt Option C ? BOTH, strictly sequenced ? which is essentially a
sharpened version of the project's existing 3-generation blueprint with two course-corrections
justified by the fresh evidence. Concretely: (1) START NOW with the specialized, sport-agnostic
TAL head (ActionFormer/ASFormer-class, MIT) over the own feature stack as the QUALITY-FIRST
rally/highlight extractor ? it is the only path that credibly reaches very-high P/R at strict
boundaries on a $0 local budget, and it's already de-risked on this corpus (AUC 0.84-0.91, LOVO
0.615 at n=3). It triples as extractor, confidence-gated label-amplifier/teacher, and the honest
baseline the VLM must beat. (2) DESIGN THE PER-VIDEO EVENT-INDEX/BOOKKEEPING FRAMEWORK NOW ?
this is no-regret: the conversational-QA loop needs durable per-video state (rally inventory,
[start,end], shot counts, fault/event tags, derived-clip refs) regardless of whether QA is a VLM
or a retrieval layer. Make the index schema a first-class data contract alongside the labeled-
corpus spine (M0.1). (3) PUNT the end-to-end conversational VLM. Evidence is decisive that a
fine-tuned VLM does not hit very-high P/R at strict temporal boundaries (best-in-class
R@0.7?50%), so folding detection+QA into one model would REGRESS the core quality metric the
product sells. Build conversational QA, WHEN it ships, as an LLM/agent doing RAG over the head's
event index (AVAS-style event-knowledge-graph) ? the 2025-26 best practice for long-video
contextual QA, and deterministic for 'make a clip of all service faults.' (4) Keep a fine-tuned
VLM as a LATER, OPTIONAL upgrade for the grounding/QA stage once the corpus is large (the proven
recipe is ~13K SFT + ~18K RL examples ? and note the public grounding-CoT data is Gemini-
generated and thus forbidden here, so you must originate that supervision, which raises the
bar). If/when adopted, prefer the two-stage pattern (VLM coarse-proposes, the head/decoder
refines boundaries ? TimeRefine/perception-decoder), not wholesale replacement. (5) On
MULTISPORT: a single sport-agnostic model is feasible and beneficial ? RacketVision (AAAI'26)
shows unified badminton+tennis+TT training improves generalization +15-19% mAP at a modest
pinpoint-precision cost; keep one shared head with sport as a feature/calibration. The binding
multisport blocker is not the model ? it's clean per-sport trajectory detectors + per-sport
labeled data, and the WASB checkpoint C3 provenance, all of which must be resolved before
commercial ship. Net: do NOT start with a VLM; do NOT skip the head; build the bookkeeping now;
treat the VLM as a deferred QA-stage option, not the quality-first extractor.",
    "confidence": "high",
    "sources": [
        "https://arxiv.org/html/2508.10922v1 ? A Survey on Video Temporal Grounding with
Multimodal LLMs (2025): VTG-MLLMs 'gradually surpassing' fine-tuned methods but 'struggle to
capture fine-grained temporal dependencies' and 'precise temporal boundary prediction'; zero-
shot Time-R1 R@0.5=60.8 R@0.7=35.3 on Charades-STA",
        "https://arxiv.org/pdf/2503.13377 ? Time-R1 (Qwen2.5-VL-7B, GRPO RL): FINE-TUNED Charades-
STA R@0.3=82.8 R@0.5=72.2 R@0.7=50.1 mIoU=60.9 ? best-class VLM still only ~50% at strict
IoU=0.7",
        "https://arxiv.org/pdf/2502.13923 ? Qwen2.5-VL Technical Report: 7B off-the-shelf
Charades-STA R@0.3=67.9 R@0.5=50.3 R@0.7=24.3 mIoU=43.6 (the project's candidate base, pre-fine-
tune)",
        "https://arxiv.org/pdf/2202.07925 ? ActionFormer (MIT): 66.8% avg mAP THUMOS14, 36.6 avg
mAP ActivityNet 1.3 ? specialized TAL boundary-precision reference",
        "https://arxiv.org/html/2512.01298 ? TBT-Former (Dec 2025): 68.0% avg mAP THUMOS14, new
SOTA, surpassing ActionFormer ? specialized heads still advancing",
        "https://arxiv.org/html/2511.17045v2 ? RacketVision (AAAI 2026, badminton+tennis+TT, 1,672
clips): unified multi-sport MS-TrackNetV3 beats single-sport +19.2% mAP tennis / +14.6%
badminton, with a modest pinpoint-precision tradeoff ? validates single sport-agnostic model",
        "https://arxiv.org/html/2412.09601v1 ? TimeRefine: VLM coarse-predict-then-refine-offsets

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two-stage; evidence the boundary-refinement/decoder pattern is needed for fine VLM
localization",
  "https://arxiv.org/html/2507.18100v1 ? TVG-R1 (EMNLP 2025 Industry): proven VLM grounding
recipe = ~13K SFT cold-start + ~18K RL; public 56K CoT labels generated by Gemini-2.5-Pro
(forbidden as a training source here)",
  "https://arxiv.org/html/2505.00254v3 ? AVAS: event-knowledge-graph index + RAG agentic
reasoning for long-video QA ? best-practice basis for the conversational layer over an event
index rather than end-to-end VLM",
  "https://arxiv.org/abs/2511.21631 ? Qwen3-VL Technical Report: explicit text-timestamp
alignment for second-level video indexing (improved grounding base, Apache-2.0)"
],
  "caveats": [
    "Benchmark transfer caveat: Charades-STA/THUMOS14/ActivityNet are open-domain. They
establish the VLM-vs-head BOUNDARY-PRECISION ceiling pattern, but absolute numbers won't map 1:1
to amateur 1080p/60fps racket footage ? rallies are more periodic/visually distinctive than
generic captioned moments, so the head's relative advantage may be even larger here (favoring
the recommendation), but this is inference, not measured.",
    "Multisport generalization of THIS feature stack is unproven: the de-risking corpus and
WASB features are badminton-only. RacketVision validates the SINGLE-MODEL concept (and a clean
MS-TrackNetV3-class detector), but tennis/TT/pickleball/padel each need clean-license trajectory
detectors + labeled data not yet in hand. Pickleball/padel have far less public dataset/detector
coverage than badminton/tennis/TT.",
    "WASB checkpoint C3 provenance remains an unresolved commercial-cleanliness blocker that
gates BOTH options (head features and pseudo-labels) ? own-train weights or swap to a clean
detector before commercial ship; a 'clean' model trained on provenance-unclear features does not
launder the provenance.",
    "The forbidden-paid-LLM-teacher constraint materially raises the VLM path's cost: the
entire public grounding-CoT data ecosystem (TVG-R1, etc.) is built on Gemini/GPT-generated CoT,
which this project bans ? so the VLM path requires originating that supervision, a non-trivial
cost that further justifies deferring it.",
    "'Very high precision/recall' should be defined operationally before the ship gate:
current best cross-video LOVO ~0.615 (n=3, linear, threshold-transfer partly unsolved) is far
from 'very high.' The path to high P/R is more data (n?dozens of matches across sports) + the
segment-level TAL head, not a base-model swap. Set the target IoU threshold explicitly (e.g.
F1@tIoU=0.5 for highlights, tighter for precise rally cuts).",
    "Qwen3-VL's grounding improvements over Qwen2.5-VL are claimed (explicit text-timestamp
alignment) but not yet quantified on Charades-STA in public sources as of mid-2026 ? re-verify
before any VLM-base commitment.",
    "Sequencing-discipline caveat per CLAUDE.md: this is a strategy recommendation; the
committed next slice is the async job model (single-tenant). Building the event-index schema now
is the no-regret action, but heavy VLM training stays gated on scaffolding + explicit owner go."
  ],
},
{
  "dimension": "Conversational Video-QA Architecture",
  "summary": "Extract once into a per-video event store; answer with a cheap query layer.
Queries are structured-numeric (text-to-SQL + clip-cut), not semantic; most of the store exists.
Build the bookkeeping now; punt the model until quality is solid.",
  "recommendation": "Adopt A: event store + session tables + a structured-query contract now;
punt the NL parser/agent until quality beats ~0.583; reject B; Gemini only a fallback.",
  "confidence": "high",
  "options": [
    {
      "name": "A event store + query layer (RECOMMENDED)",
      "what": "extract once into play_segments+events; answer with SQL + VideoCompiler clip-
cut; LLM edge-only",
      "license": "clean: SQLite/ffmpeg/MIT LLM",
      "pros": [
        "process once query many; cheap; SQL exact where VLMs hallucinate; reuses existing
tables"
      ],
    },
    {
      "cons": [
        "extraction quality is the ceiling; open-ended needs a VLM fallback"
      ],
      "fit": "strong"
    }
  ]
}

```

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    },
    {
      "name": "B end-to-end VLM (rejected)",
      "what": "one model answers each turn; no store",
      "license": "paid/forbidden-as-trainer",
      "pros": [
        "simplest; open-ended native"
      ],
      "cons": [
        "expensive; bad at counting; license-hostile"
      ],
      "fit": "weak"
    },
    {
      "name": "C RAG over captions",
      "what": "captions to vector/graph DB; Graph-RAG",
      "license": "paid-model risk",
      "pros": [
        "best for open-ended Qs"
      ],
      "cons": [
        "cannot count exactly; future wing only"
      ],
      "fit": "medium"
    }
  ],
  "sources": [
    "NVIDIA AVAS",
    "VideoAgent",
    "AVA 2505.00254",
    "Google ADK",
    "database.py",
    "CODE_MAP.md"
  ],
  "caveats": [
    "quality gates the feature (~0.583 LOVO)",
    "multisport semantics unvalidated",
    "single-tenant, owner-gated; fix the query contract early"
  ]
},
{
  "dimension": "Implementation language for the upcoming work (ML training+eval / platform / eventual on-device), for a solo Python+FastAPI founder",
  "summary": "Recommendation: stay Python everywhere for the foreseeable roadmap, and treat \"another language\" as a deferred, layer-local deployment detail ? never a rewrite. The three layers each independently point to Python or to a language choice that is structurally decoupled from your dev language, so there is no layer where a second language earns its keep against solo-founder velocity right now.\n\n(1) ML TRAINING+EVAL is effectively Python-locked and that is fine: the PyTorch/Transformers/ms-swift/ActionFormer/ASFormer stack is Python, and ms-swift is invoked as a CLI (`swift sft/eval`) ? so even the orchestrator just shells out to a subprocess. There is no non-Python alternative that retains native-video VLM training (the doc's decisive finding) or your dependency-free numpy LogisticRegression Gen-0 seed. Locking here costs nothing because training is batch/offline, not a latency hot path.\n\n(2) PLATFORM (async jobs, multi-tenant API, storage/compute backends) has no real Python \"hot path\" to escape. The heavy bytes/GPU work already runs out-of-process (ffmpeg Windows-side, WASB in WSL, cloud GPU per-job via Modal/RunPod) ? Python is the conductor, not the orchestra. The control plane is I/O-bound (DB, signed URLs, job status, polling), exactly where the GIL never bound you and where Python 3.14 free-threading + asyncio now also remove the residual CPU-bound penalty. FastAPI 0.136+ officially supports free-threaded Python. fsspec/smart_open/Arq/Hatchet (all Python, all named in PLATFORM_ARCHITECTURE.md) cover storage+queue. A Go/Rust hot-path service would buy single-digit-ms wins on requests dwarfed by 40-minute GPU jobs ? pure overhead for a solo founder.\n\n(3) ON-DEVICE is the one layer where another language eventually appears ? but it is DEPLOYMENT-only and structurally decoupled from your dev language by the export boundary. You train in Python, export to ONNX/Core ML/ExecuTorch/GGUF, and the on-device host (Swift for iOS Core ML, C++/Kotlin for ExecuTorch/Android, Rust via the `ort` crate) runs the artifact with

```

zero Python dependency. ONNX Runtime alone exposes C++/C#/Java/Rust/JS APIs over the same model. So on-device does NOT dictate the dev language; it dictates a thin, separate, much-later runtime shell you write only if/when you ship a mobile compressed model. Per the owner's reframe, on-device is explicitly LATER (an upload-cost saver), so this is the correct layer to punt ? and punting it costs nothing architecturally because the export boundary is the seam.\n\nCross-cutting point on the conversational-QA north star and per-video bookkeeping: this REINFORCES Python-everywhere. The bookkeeping (rally inventory, shot counts, fault tags, timestamps, derived clips) is a durable relational store + retrieval/tool layer ? SQLite?Postgres + a thin query/agent layer ? which is bread-and-butter FastAPI/Python. Splitting language here would fracture the one codebase that has to hold per-video state, the eval harness, and the provider registry together.\n\nWhen a second language IS justified (write it down, don't build it now): only at on-device ship time, and only the runtime shell (Swift/Kotlin/C++/Rust host around an exported artifact) ? never the trainer, never the control plane. Premature triggers to resist: rewriting the job queue in Go \"for throughput,\" or a Rust trajectory/feature-extraction service ? both are out-of-process or numpy-vectorized already; reach for them only if profiling on real load shows a Python CPU hot path that free-threading + a C-extension (numpy/Cython/PyO3-as-a-module) can't fix.",

```
"options": [
  {
    "name": "Python everywhere (all 3 layers), on-device runtime shell deferred",
    "what": "Keep FastAPI control plane, Python ML training/eval (PyTorch + ms-swift CLI + ActionFormer/ASFormer), Python platform (fsspec/smart_open storage, Arq/Hatchet jobs, Postgres + repository layer). The eventual on-device model is a Python-trained artifact exported to ONNX/Core ML/ExecuTorch; the device-side host language is chosen ONLY at mobile-ship time and is a thin, separate shell.",
    "license": "All tooling MIT/Apache/BSD-compatible (FastAPI MIT, ms-swift Apache-2.0, ActionFormer/ASFormer MIT, fsspec/smart_open BSD/MIT, Arq MIT, ONNX Runtime MIT) ? clean under the never-corrupted guardrail.",
    "pros": [
      "Maximum solo-founder velocity: one language, one toolchain, one mental model across training, eval, platform, and per-video bookkeeping",
      "ML stack has no real non-Python option that keeps native-video VLM training (ms-swift) and your existing numpy LogReg Gen-0 seed ? staying Python avoids re-implementing the de-risking artifacts",
      "Conversational-QA bookkeeping (durable per-video state + retrieval/tool layer) is exactly FastAPI/Python's strength; keeps state, eval harness, and provider registry in one codebase",
      "Python 3.14 free-threading + asyncio + FastAPI 0.136 free-threaded wheels remove the historical GIL objection for CPU-bound endpoints if it ever bites",
      "Heavy work already runs out-of-process (ffmpeg, WSL WASB, cloud GPU) ? Python is the conductor, so 'hot path in Python' is largely a non-issue",
      "On-device export boundary (ONNX/Core ML/GGUF) means the device language is a future deployment detail, not a dev-language commitment now"
    ],
    "cons": [
      "On-device, when it ships, still requires writing SOME non-Python runtime shell (Swift/Kotlin/C++/Rust) ? Python doesn't run well on phones; this is unavoidable regardless of dev language",
      "Free-threaded Python is new (3.13 had ~40% single-thread penalty; 3.14 cut it to 5-10%) ? production maturity still settling, so treat as upside not dependency",
      "Python packaging/deploy ergonomics for a self-hosted serving stack are heavier than a single Go/Rust binary (mitigated by containers + Modal-per-job)"
    ],
    "fit": "strong"
  },
  {
    "name": "Python core + Go/Rust hot-path service for the platform control plane / job dispatch",
    "what": "Keep Python for ML + API surface, but extract job-dispatch / high-concurrency endpoints / a trajectory-feature microservice into Go or Rust 'for throughput and a single static binary.'",
    "license": "Go (BSD), Rust (MIT/Apache dual) ? both clean. ONNX `ort` Rust crate is MIT/Apache. No license blocker; the objection is purely cost/benefit.",
    "pros": [
      "Single static binary deploy + lower memory footprint for the always-on control
```

```

plane",
    "Rust `ort` could serve an exported ONNX model with no Python in the serving path, if
a self-hosted inference tier ever needs maximal density",
    "Genuinely faster per-request latency on CPU-bound control-plane work"
],
"cons": [
    "Premature for a solo founder: control plane is I/O-bound (DB, signed URLs, polling)
where Python was never bottlenecked; wins are single-digit-ms against 40-min GPU jobs",
    "Two languages = two toolchains, two CI paths, FFI/serialization seams, fractured per-
video-QA state logic ? directly taxes the velocity the owner needs",
    "No current Python hot path identified to justify it; heavy compute is already out-of-
process or numpy-vectorized",
    "If a Python CPU hot path ever appears, cheaper fixes exist first (free-threading,
numpy/Cython/Py03 C-extension) before standing up a second service"
],
"fit": "weak"
},
{
    "name": "Rust/C++ on-device runtime shell at mobile-ship time (deferred, layer-local)",
    "what": "When the LATER compression goal ships a phone-side model, write only the
device-side inference host in the platform-native language: Swift+Core ML (iOS), Kotlin/C++
+ExecuTorch (Android), or Rust `ort`/C++ for a desktop power-user binary ? wrapping a Python-
trained, exported artifact.",
    "license": "ExecuTorch (BSD), Core ML (Apple SDK, app-distribution clean), ONNX Runtime
(MIT), llama.cpp (MIT). Clean ? but verify the BASE model's on-device license (note: Gemma 3
270M GGUF is commonly cited for on-device yet is license-REJECTED per your guardrail; stay on
Qwen3-VL-4B Apache-2.0 lineage).",
    "pros": [
        "Correct, unavoidable place for a second language: phones can't run the Python
trainer; the device needs a native host",
        "Fully decoupled by the export boundary ? does not touch training or control-plane
code, so it never fractures the main codebase",
        "Can be deferred with zero architectural cost today; the ONNX/Core ML export step is
the seam, planned for but not built now"
    ],
    "cons": [
        "Real engineering surface (mobile build, quantization validation, KV-cache mgmt) ? but
only when on-device is actually prioritized, which the owner explicitly defers",
        "Adds a maintenance target (app + model-version compatibility) ? budget it as a
distinct later phase, not now",
        "Ties up scarce solo-founder time the moment it starts; gate it behind proven quality
+ a measured upload-cost ROI"
    ],
    "fit": "medium"
}
],
"recommendation": "Stay Python everywhere now, and explicitly write down \"another
language\" as a deferred, layer-local DEPLOYMENT decision ? not a rewrite and not a hot-path
split. Concretely: (a) ML training+eval stays Python (PyTorch + ms-swift CLI as a subprocess +
ActionFormer/ASFormer + your numpy LogReg Gen-0 seed); it is Python-locked and that costs
nothing because training is offline/batch. (b) Platform stays Python/FastAPI; do NOT introduce
Go/Rust for the control plane ? it is I/O-bound, the heavy work is already out-of-process
(ffmpeg/WSL/cloud GPU), and Python 3.14 free-threading + FastAPI 0.136 cover the only remaining
CPU objection. Use the Python-native pieces already chosen in PLATFORM_ARCHITECTURE.md
(fsspec/smart_open, Arq?Hatchet, Postgres + repository layer). (c) On-device is the ONLY layer
where a second language eventually appears, and it does NOT dictate your dev language: you train
in Python and export to ONNX/Core ML/ExecuTorch/GGUF across the export boundary, then write a
thin native runtime shell (Swift/Kotlin/C++/Rust-`ort`) ONLY when the LATER upload-cost-saver
model actually ships. Until then, design the seam (an exportable model + a
provider/ComputeBackend registry slot) and punt the shell. This keeps the conversational-QA per-
video bookkeeping, the eval harness, and the provider registry in one Python codebase ? which is
exactly what the quality-first, multisport, stateful-QA reframe needs. The single concrete
language action now is to ensure the model architecture you train is export-clean (ONNX/Core ML-
friendly ops) so on-device stays reachable without re-architecting ? a constraint on the model,
not the dev language.",

```

```

"confidence": "high",
"sources": [
  "docs/PLATFORM_ARCHITECTURE.md (§3.2 ComputeBackend, §5 orchestration
  Arq/Hatchet/Temporal, §5A ms-swift training forge, fsspec/smart_open storage) ? existing Python-
  native posture",
  "docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md (Gen-0 ActionFormer/ASFormer MIT on local 2070,
  Gen-2 Qwen3-VL-4B Apache-2.0, Modal/RunPod cloud-per-job)",
  "docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md (ms-swift 'only native-video'
  finding; on-device VLM unbenchmarked; numpy LogReg Gen-0 seed)",
  "docs/CODE_MAP.md (heavy compute already out-of-process: ffmpeg Windows-side, WASB in
  WSL2; dependency-free numpy classifier.LogisticRegression; FastAPI app in backend/main.py)",
  "https://github.com/modelscope/ms-swift ? ms-swift is Python + CLI (swift sft/rlhf/eval),
  Apache-2.0, native video+audio training, 400+ MLLMs",
  "https://pypi.org/project/ms-swift/ ? ms-swift CLI/Python invocation surface",
  "https://onnxruntime.ai/docs/ ? ONNX Runtime cross-language APIs (C++/C#/Java/Rust via
  ort/JS): train in Python, run model with no Python dependency",
  "https://docs.rs/onnxruntime/ and ort crate ? Rust on-device serving of exported ONNX
  (YOLO-in-Rust example: Python-trained, Rust-served)",
  "https://www.edge-ai-vision.com/2026/01/on-device-llms-in-2026-what-changed-what-matters-
  whats-next/ ? ExecuTorch 1.0 GA (mobile), llama.cpp (desktop/proto), MLX (Apple); on-device
  runtime is a deployment-language choice, decoupled from training",
  "https://edgeaistack.ai/blog/edge-llm-runtime-stack-2026/ ? edge runtime stack
  (llama.cpp/ExecuTorch/MLX/vLLM) comparison; export-then-run model",
  "https://medium.com/@kevaldekivadiya2415/python-gil-vs-no-gil-real-fastapi-benchmarks-
  with-free-threaded-python-3-13-... ? FastAPI free-threaded benchmarks: CPU-bound ~8x up,
  I/O-bound unchanged (GIL never bound I/O)",
  "https://blog.imseankim.com/python-3-14-free-threading-jit-compiler-gil-removal-2026/ ?
  Python 3.14 cut free-threaded single-thread penalty to 5-10%, lock-free asyncio; FastAPI 0.136
  free-threaded support"
],
"caveats": [
  "The on-device second-language need is REAL but DEFERRED and layer-local: when a phone
  model ships you WILL write some Swift/Kotlin/C++/Rust runtime shell ? Python doesn't run on-
  device. The recommendation is to defer it via the export boundary, not to claim it never
  happens.",
  "Free-threaded Python (3.13t/3.14t) is the load-bearing reason the GIL objection is moot
  for CPU-bound endpoints; it is new and still maturing in production. Treat it as upside, not a
  hard dependency ? the platform is I/O-bound and fine on standard CPython regardless.",
  "One concrete language-adjacent constraint flows onto the MODEL (not the dev language):
  train with export-clean / ONNX/Core ML-friendly ops so the eventual compression target stays
  reachable. The tension that the immediate high-quality model isn't 8GB-constrained but must
  remain later-compressible is a model-architecture choice, handled in the model dimension, not
  here.",
  "An on-device base-model license note surfaced incidentally: Gemma 3 270M GGUF is
  frequently cited for on-device but is REJECTED by your guardrail (viral terms). Keep on-device
  bases on the Qwen3-VL-4B / Gemma-4 Apache-2.0 lineage ? a licensing item for the model
  dimension, flagged so the convenient on-device default isn't accidentally adopted.",
  "If profiling on real multi-tenant load ever shows a genuine Python CPU hot path in the
  control plane, escalate in this order BEFORE a second language: free-threading ->
  numpy/vectorize -> a C-extension via Cython/PyO3 (still a Python module) -> only then a separate
  Go/Rust service. No such hot path is identified today.",
  "ms-swift being Python-locked is not a risk: it is invoked as a CLI subprocess, so even a
  non-Python orchestrator would shell out identically ? the lock-in is to the PyTorch ecosystem,
  which is industry-standard and unavoidable for VLM fine-tuning."
]
}
]

```

2. assistant (2026-06-09T11:39:56.098Z)

I'll verify the load-bearing claims in these findings against primary sources. Let me start by loading the web tools and reading the relevant project docs.

3. assistant (2026-06-09T11:40:00.233Z)

Let me read the project docs referenced in the findings, then verify the web claims in parallel.

4. user (2026-06-09T11:40:01.308Z)

```
1      # Owned-model training study (rally segmentation ? end-to-end multimodal)
2
3      **Date:** 2026-06-08 . **STATUS: DESIGN BLUEPRINT (no implementation until PLATFORM
P0?P4 + healthy corpus; owner sign-off required).**
4
5      > How to train our **OWNED** rally/highlight model **reusing the videos we label**,
aligned to the
6      > Apache-2.0 / no-paid-LLM-training / privacy-local guardrails. Produced by an 11-agent
research
7      > workflow (ground ? 5 web-research dimensions ? adversarial verification of
costs/licenses/privacy ?
8      > synthesis + completeness critic; 35 sources). Empirical results from THIS session
(LOVO + a learned
9      > prototype) are folded in and **resolve the study's biggest open unknown** ? see $"What
we proved today".
10     > Numbers from web sources are 2026 snapshots in a  $\pm 15\%$  band; **re-quote live before
budgeting.**
11
12     ## Thesis
13
14     Train the owned model in **three staged generations off ONE model-agnostic labeled
corpus** ? a label
15     collected once feeds every generation, so labels are never re-collected. **The moat is
the consented
16     dataset + the eval harness, never the weights.**
17
18     | Gen | When | Approach | Base / model | Compute |
19     |---|---|---|---|---|
20     | **Gen-0** | now ? unblocks at n?5 matches | tiny **temporal action segmentation
(TAS)** head over a hand-built per-frame feature stack from the WASB trajectory we already
produce; per-frame rally(1)/non-rally(0) ? merge into [start,end] | **ASFormer (MIT, ~1.1M
params)** primary; **ActionFormer (MIT)** A/B challenger (overtakes once corpus = hundreds of
matches) | **local RTX 2070 Super**, trains in minutes, $0 |
21     | **Gen-1 / Option A** | 6?12 mo | concatenate more local cue channels onto the SAME
head: trajectory + audio-onset + pose/court serve-gating; fusion in our rally layer, cues
swappable | same ASFormer/ActionFormer, wider input | local + occasional cloud burst (pseudo-
labeling) |
22     | **Gen-2 / Option B** | 12?24+ mo, after scaffolding + healthy corpus | end-to-end
multimodal VLM: (frames+audio+pose+trajectory) ? rally [start,end] (+semantics); SFT cold-start
then **GRPO/RFT with a temporal-IoU-vs-golden reward** | **Qwen3-VL-4B-Instruct** (Apache-2.0,
on-device) primary; **Qwen2.5-VL-7B** fallback; **Gemma 4 E4B/12B-Unified** (Apache-2.0 + native
video+audio) end-state | rented 24?48 GB A100/L40S (does NOT fit 8 GB box) |
23
24     ## What we proved today (de-risks the blueprint)
25
26     The study's ***"dominant empirical unknown"*** was: *can a low-dim WASB trajectory (?3?4
raw dims) feed a
27     TAS head, given ASFormer's small-data result used 2048-d I3D features?* This session
answered it.
28
29     - **Cross-video LOVO of the current heuristic** (`calibrate_local`, Adarsh-96 +
Kushagra-32, human labels):
30     mean held-out F1 **0.443  $\pm$  0.279**; Adarsh 0.722, **Kushagra 0.163** (its own ceiling
? *no* velocity-windowing
31     config does better). **The  $\pm 0.279$  spread is method variance, not config.** ? This also
**corrects the study's
32     "~0.73 bar"**: that was a fit-on-self single-match number; the honest cross-video
heuristic bar is **~0.44**,
33     which *lowers* the Gen-0 ship gate.
34     - **A learned per-frame prototype** (`backend/eval/rally_seq_proto.py`: 6 hand-built
features ? visibility rate,
```

35 mean/max shuttle speed, ****net-crossing rate****, x/y spread ? into the repo's
 dependency-free LogisticRegression,
 36 honest LOVO): ****per-frame AUC 0.878 (Adarsh) / 0.841 (Kushagra), held-out.**** On
 Kushagra ? the match the
 37 heuristic ***cannot*** segment (0.163) ? a calibrated threshold reaches ****segment F1 0.578**
 (3.5×)**. The naive
 38 cross-video threshold gives only 0.049, so ****the remaining gap is threshold/operating-**
 point calibration at n=2,
 39 not representation.**
 40
 41 - ****n=3 update (doubles match added ? `LargeTestVideo`, 49 rallies):**** the learned
 prototype ****overtakes the**
 42 heuristic.** Mean held-out LOVO F1 ? learned ****0.377 ? 0.615****, heuristic ****0.443 ?**
 0.544**. Kushagra's learned
 43 held-out F1 jumped ****0.049 ? 0.417**** (one added match closed most of the calibration
 gap); the ****doubles**** match,
 44 held out and trained only on the two singles, scored AUC ****0.914**** / F1 0.729
 (generalizes across play type +
 45 frame rate). The heuristic still cannot rescue Kushagra at any n (0.151, capped).
****The honest "bar to beat" is**
 46 thus ~0.54 (n=3), not 0.73 ? and the learned path already clears it on the mean and
 keeps rising with n.**
 47
 48 ****Conclusion:**** the rally signal IS learnable from the WASB trajectory features, cross-
 footage (AUC ~0.85?0.91) ?
 49 the heuristic just can't exploit it. This ****validates the Gen-0 ASFormer-over-WASB-**
 features path**, shows the
 50 hand-built feature stack the study calls for is the right lever, and confirms the
 bottleneck is ****data + per-video**
 51 calibration (n?5 matches), not method.** The prototype is the seed of Gen-0, and the
 n=2?n=3 trend (learned mean
 52 +0.24, overtaking the heuristic) is the empirical green light.
 53
 54 **## Platform ranking (privacy + cost lens; the fine-tune is a sub-24h, \$10?50 job ?**
 privacy/ergonomics dominate)
 55
 56 1. ****Modal ? PRIMARY.**** Strongest ***written*** no-access contract (verbatim: "Modal will
 never access? the inputs/outputs to your Functions, or any data you store"), ?7-day TTL,
 SOC2-II, HIPAA-BAA; per-second scale-to-zero matches retrain-on-each-batch. A100 80 GB
 ~\$2.50/hr, H100 ~\$3.95/hr (corrected down from cited). Privacy is the load-bearing reason, not
 price.
 57 2. ****RunPod (Secure Cloud tier ONLY) ? FALLBACK / Option-B scale-up.**** Cheapest vetted-
 DC self-serve; A100 80 GB ~\$1.39?1.49/hr. Community/P2P tier is UNVETTED ? ****never**** touches
 consented video. Confirm AES-256/DPA at booking.
 58 3. ****Vertex AI / GCP ? ON BENCH.**** Only strong-privacy option with an India region, but
 it's ****capacity-gated**** (asia-south1 high-end GPU is invitation-only, H100-class A3 not A100)
 and ****tuning Gemini yields NO portable on-device weights**** ? only tuning an Apache-2.0 base in
 Model Garden does. Use only if India residency-at-rest is mandated.
 59 4. ****Self-hosted RTX 2070 Super 8 GB**** ? Gen-0 training + ALL inference + CI only.
****Cannot**** QLoRA a 7B video VLM (~20?24 GB). Best privacy (nothing leaves the box).
 60 5. ****Fireworks**** (managed, contractual no-train + ZDR) usable; ****Together BLOCKED****
 until a signed DPA + explicit no-train clause. AWS SageMaker / Azure ML = heavy for a solo
 founder; India-residency only.
 61
 62 **## Label reuse ? one corpus, three transcoders (the actual moat)**
 63
 64 Promote the rally CSVs from eval-only files to ****one model-agnostic corpus****: JSONL, one
 row per
 65 `(video\_id, rally\_ordinal)` = `{start, end, ending\_reason, sport, source`
 (human|pseudo|vendor),
 66 annotator/provenance, consent/opt-in, label_type, split (BY MATCH not clip),
 trajectory_ref}`. The
 67 provenance/consent fields are the ****enforcement point**** for the legal guardrails. Three
 deterministic
 68 transcoders read it, zero re-labeling:

```

69     1. **Heuristic/LogReg view** ? already exists
(`backend/eval/training.py::label_candidates` via `metrics.match_intervals`).
70     2. **TAL view** ? ActivityNet JSON (`segment:[start,end]`, `label:'rally'`); our seconds
map 1:1.
71     3. **VLM clip?QA view** ? "list every rally [start,end] in this clip"; `ending_reason` ?
a second QA task.
72
73     **One scorer spine:** reuse `metrics.match_intervals` for heuristic-labeling, TAL eval,
AND the Gen-2 RFT
74     reward ? no metric drift. **Leakage guard:** split **by match, never by clip**.
**Amplification (all
75     guardrail-clean):** open-teacher pseudo-labels (WASB/own-model, confidence-gated ? never
paid-LLM),
76     boundary-preserving temporal augmentation, active learning to spend the Roora budget on
highest-uncertainty
77     matches. ? **The new partial-label windowing**
(`eval_partial_labels.restrict_to_window`) must flow into the
78     TAL/VLM views so prefix-labeled videos aren't mis-scored (a corpus that's partially
tagged is the norm).
79
80     ## Licensing & privacy guardrails (all respected; verify per-checkpoint)
81
82     - **Bases: Apache-2.0/MIT only.** APPROVED: Qwen2.5-VL-7B/32B (Apache-2.0),
Qwen3-VL-4B/8B (Apache-2.0),
83     Gemma 4 E2B/E4B/12B-Unified (Apache-2.0 + native video/audio), InternVL3 ?38B
(MIT/Apache), ASFormer/
84     ActionFormer/TriDet (MIT). **REJECTED:** Gemma 3 (viral Gemma Terms + PUP),
Qwen2.5-VL-3B (NC research),
85     Qwen-72B/InternVL3-78B (100M-MAU cap), Llama Vision (700M-MAU), DAMO VideoLLaMA3
weights (NC + paid-LLM-trained),
86     TimeChat/VTimeLLM/Momentor (NC/no-license = method-reference only), VideoMAE weights
(CC-BY-NC), `yabufarha/ms-tcn`
87     (Commons Clause ? use `sj-li/MS-TCN2` MIT). ? **Gemma 4 size names =
E2B/E4B/26B-A4B/31B/12B-Unified ? there is no plain "4B".**
88     - **No paid-LLM training data** ? Gemini/OpenAI/Anthropic = inference/eval/teacher-
reference only, **never** a
89     training signal. Gemini silver labels stay out of every transcoder.
90     - **Exclude minors by default + separate training opt-in;** per-user adapters trained
only on that user's
91     consented video (deletion = drop the adapter). Enforced at the corpus-membership layer
before transcoding.
92     - **Privacy lever = local inference** (Gen-0 ~1.1M params; Gen-2 4B quantized). Any
cloud *training* needs a
93     signed DPA + no-train assurance; route only owned/consented footage, never user
uploads.
94
95     ## Cost (spend on labels, not GPUs)
96
97     - **Gen-0:** ~$0 (local 2070 Super; trains in minutes; ~4.5 GB VRAM).
98     - **Gen-2 LoRA:** ~$5?40/run on a rented 24?48 GB GPU (RunPod cheapest; **VRAM/token
cost on real 1080p/60fps
99     clips is UNVERIFIED ? 24 GB is a floor, re-benchmark**). Managed (Fireworks/Together)
~$2?30 *if* text-priced
100     (video tokens can balloon). RFT = compute-heavy (multiple rollouts/sample).
101     - **Inference crossover (the unanchored number):** self-hosted 7B beats Gemini
(~$0.05/10-min clip) **only**
102     above sustained QA volume; below it, **paid Gemini is cheaper** ? paid-primary now,
owned-primary later.
103     **Benchmark $/call before committing** (COMMERCIALIZATION C7).
104
105     ## Open risks & critic caveats (read before acting)
106
107     - **?? WASB checkpoint provenance (C3) propagates.** The study calls WASB "MIT," but
that's the *code*; the
108     distributed `wasb_badminton_best.pth.tar` is **academic-data-trained, provenance

```

```

unresolved** (COMMERCIALIZATION
109      C3, still open). Every Gen-0/1 plan uses WASB features as input AND as a pseudo-label
teacher ? a "clean" owned
110      model trained on provenance-unclear features **does not launder the provenance.**
ROADMAP Phase-4 "train own WASB
111      weights" is a **prerequisite, not an aside.**
112      - **Gemma 4 "clean + native-video" rests on post-cutoff secondary sources** ? keep AMBER
until a raw HF LICENSE
113      revision is verified, given the Gemma-3 rejection. Its timestamp-emission recipe is
less documented than Qwen's.
114      - **Existing distilled model was trained on Gemini silver labels** ? the (correct) eval-
only policy now **forbids
115      this; that pipeline must be purged of Gemini-derived training labels** to comply.
116      - **Policy ratifications needed:** guardrail literally says "Apache-2.0 only" but the
approved list includes MIT ?
117      ratify "Apache-2.0/MIT". The MAU-cap rejections (72B/78B) may be over-conservative at
our scale.
118      - **On-device privacy claim is unbenchmarked** ? no tokens/sec for a quantized 4B VLM on
8 GB Turing at
119      1080p/60fps. **Measure before relying on it.**
120      - **Amplification is load-bearing for Gen-2** ? 128 raw rallies can't reach the ~5k?20k
clip?QA band alone; if
121      augmentation + QA-multiplexing + pseudo-labels underdeliver, Option B is under-fed.
Amateur 60fps may need more
122      labels than broadcast-trained literature implies.
123      - **Dropped option worth keeping:** InternVideo2 encoder (MIT) + light grounding head ?
the *only* data-realistic
124      owned-video experiment available at n?128 today. And: a **one-time ~$1?2k 24 GB local
card** (e.g. used 3090)
125      would preserve the privacy lever for *training* and dissolve most of the cloud-DPA
analysis ? uncoded here.
126
127      ## Open questions (need the GPU/WSL machine-side suite ? not runnable in this dev
container)
128
129      1. Does ASFormer/ActionFormer over the WASB feature stack beat the **honest ~0.44 LOVO**
bar at n?5 matches?
130      (Today's prototype says the per-frame AUC is there ? 0.84 ? so this is now *likely*;
confirm at segment level.)
131      2. Real VRAM + token cost of fine-tuning Qwen3-VL-4B / Qwen2.5-VL-7B on our actual
clip?QA frame counts?
132      3. Working Gemma-4 timestamp-emission recipe + exact checkpoint (E4B vs 12B-Unified)?
133      4. Measured self-hosted-7B $/call + latency vs Gemini on the target workload (the owned-
model ROI crossover)?
134      5. Will Together/Lambda/Azure sign a DPA with explicit no-train for consented video,
India-configurable?
135      6. Realistic clip?QA yield from 128 rallies via augmentation + QA-multiplexing + pseudo-
labels?
136
137      ## Sequencing (owner directive, non-negotiable)
138
139      This is a **design blueprint only.** It unblocks PLATFORM **P5** but starts **no
implementation** until P0?P4
140      scaffolding (async jobs, storage/compute backends) is rock-solid AND the golden corpus
is healthy. Paid Gemini
141      stays the always-ready primary/fallback. The only items safe to act on now ? **and even
these need owner
142      sign-off before code** ? are (a) the model-agnostic corpus format and (b) growing n?2 ?
n?5 matches (a small
143      Roora batch), which unblocks honest tuning immediately and is the **highest-ROI spend
(labels, not GPUs).**
144
145      *Full agent transcript + 35 sources: workflow `wf_49aab888-e10` (this session).*
146

```

5. user (2026-06-09T11:40:08.989Z)

Web search results for query: "Time-R1 Qwen2.5-VL-7B Charades-STA R@0.7 mIoU temporal grounding GRPO"

Links: [{"title": "Invert4TVG: A Temporal Video Grounding Framework with Inversion Tasks Preserving Action Understanding Ability", "url": "https://arxiv.org/html/2508.07388"}, {"title": "Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding", "url": "https://papers.neurips.cc/paper_files/paper/2025/file/7801b29c93b599b8d0c44138596bdeed-Paper-Conference.pdf"}, {"title": "Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding", "url": "https://arxiv.org/html/2503.13377v2"}, {"title": "Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding", "url": "https://arxiv.org/pdf/2503.13377"}, {"title": "TempSamp-R1: Effective Temporal Sampling with ...", "url": "https://openreview.net/pdf/2d01d96582304da4be0ebb7741e61b36c9a8d81d.pdf"}, {"title": "Datasets and Recipes for Video Temporal Grounding via Reinforcement Learning", "url": "https://arxiv.org/pdf/2507.18100"}, {"title": "Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding", "url": "https://xuboshen.github.io/Time-R1/"}, {"title": "TimeLens: Rethinking Video Temporal Grounding with Multimodal LLMs", "url": "https://arxiv.org/pdf/2512.14698"}]

Based on the search results, I found relevant information about Time-R1, Qwen2.5-VL-7B, and temporal grounding performance:

Key Findings:

Time-R1 is built on the Qwen2.5-VL-7B base model. On the Charades-STA dataset, Time-R1's R1@0.7 score reaches 50.1 and its R1@0.5 score reaches 72.2.

GRPO (Group Relative Policy Optimization) is a reinforcement learning algorithm proposed by DeepSeek-AI, and Time-R1 fine-tunes LVLs (specifically the Qwen2.5-VL model series) by integrating both temporal video grounding (TVG) and inverse-TVG tasks using GRPO.

The "R1@m" evaluation protocol computes the percentage of samples where the top-1 predicted segment has an IoU greater than a threshold m, with m ∈ {0.3, 0.5, 0.7}, and mIoU calculates the average IoU on all testing data as an alternative metric.

The search results show that on Charades-STA, the 7B variant of Invert4TVG achieves an R1@0.7 of 51.4, exceeding Time-R1 (50.1), indicating that Time-R1's performance on this benchmark is competitive among recent models.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-09T11:40:14.044Z)

The Time-R1 numbers are confirmed (R@0.7=50.1, R@0.5=72.2 fine-tuned on Charades-STA). Now let me verify the most load-bearing claims in parallel: Qwen3-VL's Apache-2.0 license + timestamp claim, the RacketVision multisport claim, and the Qwen2.5-VL off-the-shelf number.

7. user (2026-06-09T11:40:19.365Z)

Model License and Key Features

License: apache-2.0

Timestamp Grounding & Temporal Localization: Yes, the model card explicitly mentions this capability. The relevant section states:

"Text?Timestamp Alignment: Moves beyond T?RoPE to precise, timestamp?grounded event localization for stronger video temporal modeling."

****Long Video Context Length:**** Yes, this is highlighted as a major enhancement. The model card specifies:

"Long Context & Video Understanding: Native 256K context, expandable to 1M; handles books and hours-long video with full recall and second-level indexing."

These features represent significant architectural improvements in how the model processes temporal information and handles extended video sequences.

8. user (2026-06-09T11:40:23.672Z)

Web search results for query: "RacketVision AAAI 2026 badminton tennis table tennis unified multi-sport mAP improvement MS-TrackNetV3"

Links: [{"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis", "url": "https://arxiv.org/html/2511.17045v2"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis", "url": "https://arxiv.org/pdf/2511.17045"}, {"title": "GitHub - OrcustD/RacketVision: The official repository of the paper \"RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis\"", "url": "https://github.com/OrcustD/RacketVision/"}, {"title": "[2511.17045] RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis", "url": "https://arxiv.org/abs/2511.17045"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis | AI Research Paper Details", "url": "https://www.aimodels.fyi/papers/arxiv/racketvision-multiple-racket-sports-benchmark-unified-ball"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified", "url": "https://ojs.aaai.org/index.php/AAAI/article/view/37362/41324"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis | Proceedings of the AAAI Conference on Artificial Intelligence", "url": "https://ojs.aaai.org/index.php/AAAI/article/view/37362"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis | Request PDF", "url": "https://www.researchgate.net/publication/397895011_RacketVision_A_Multiple_Racket_Sports_Benchmark_for_Unified_Ball_and_Racket_Analysis"}, {"title": "TrackNet: A Deep Learning Network for Tracking High-speed and Tiny Objects in Sports Applications* | Request PDF", "url": "https://www.researchgate.net/publication/337538165_TrackNet_A_Deep_Learning_Network_for_Tracking_High-speed_and_Tiny_Objects_in_Sports_Applications"}]

Based on the search results, here's what I found about RacketVision and its AAAI 2026 acceptance:

Overview

RacketVision's paper was accepted by AAAI 2026 (Oral). RacketVision is a novel dataset and benchmark for advancing computer vision in sports analytics, covering table tennis, tennis, and badminton.

Dataset Details

The dataset comprises a collection of 1,672 video clips (435,179 frames, 12,755 seconds) spanning badminton, tennis, and table tennis. The dataset is the first to provide large-scale, fine-grained annotations for racket pose alongside traditional ball positions, enabling research into complex human-object interactions.

Tasks and Performance

It is designed to tackle three interconnected tasks: fine-grained ball tracking, articulated racket pose estimation, and predictive ball trajectory forecasting.

mAP Improvements with MS-TrackNetV3

The best overall results in the benchmark are achieved by MS-TrackNetV3 with 4 frames. Regarding the multi-sport improvements, by comparing the best multi-sport model (MS-TrackNetV3, #F=4) with the best single-sport model (TrackNetV3, #F=4), the MS model boosts mAP by a significant 19.2% in Tennis (81.9 vs. 68.7) and 14.6% in Badminton (83.1 vs. 72.8).

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-09T11:40:33.585Z)

Both confirmed. Qwen3-VL-8B-Instruct is apache-2.0 with explicit "Text-Timestamp Alignment" and 256K?1M context. RacketVision (AAAI'26 Oral) confirms +19.2% tennis / +14.6% badminton mAP from unified multi-sport training.

Now I need to scrutinize the most important nuance: the finding claims the VLM does NOT reach high P/R at strict boundaries (R@0.7?50%), which is the load-bearing skeptical claim. Let me verify the RacketVision claim more carefully ? it's about ball-tracking mAP, NOT temporal localization/rally start-end. This is a potential overclaim. Let me also verify the Qwen2.5-VL off-the-shelf number and check whether RacketVision actually measures temporal localization at all.

10. user (2026-06-09T11:40:40.124Z)

RacketVision Tasks and mAP Metrics

Three Core Tasks

RacketVision focuses exclusively on three tasks: **ball tracking**, **racket pose estimation**, and **ball trajectory prediction**. It does not include temporal action localization, rally segmentation, or highlight detection.

The benchmark defines these as: "ball tracking, racket pose estimation, and predictive ball trajectory forecasting."

mAP Improvement Details

The +19.2% and +14.6% mAP improvements measure **ball detection accuracy**, not temporal event localization. Specifically, the text states:

"By comparing the best multi-sport model (MS-TrackNetV3, #F=4, bold results) with the best single-sport model (TrackNetV3, #F=4, underlined results), we observe a clear trend of improved generalization. For example, the MS model boosts mAP by a significant 19.2% in Tennis (81.9 vs. 68.7) and 14.6% in Badminton (83.1 vs. 72.5)."

These gains represent improvements in detecting ball positions across frames when training on multiple sports simultaneously, rather than measuring temporal event boundaries or highlight spans.

11. user (2026-06-09T11:40:43.695Z)

Web search results for query: "Qwen2.5-VL technical report Charades-STA off-the-shelf R@0.5 50.3 R@0.7 24.3 mIoU 43.6 temporal grounding"

Links: [{"title": "Agentic Video Intelligence: A Flexible Framework for Advanced Video Exploration and Understanding", "url": "https://arxiv.org/pdf/2511.14446"}, {"title": "Qwen2.5-VL: Advanced Vision-Language Model", "url": "https://www.emergentmind.com/topics/qwen2-5-vl-model"}, {"title": "TAR-TVG: Enhancing VLMs with Timestamp Anchor-Constrained Reasoning for Temporal Video Grounding", "url": "https://arxiv.org/pdf/2508.07683"}, {"title": "TempSamp-R1: Effective Temporal Sampling with Reinforcement Fine-Tuning for Video LLMs", "url": "https://arxiv.org/pdf/2509.18056"}, {"title": "Video-in-the-Loop: Span-Grounded Long Video QA with Interleaved Reasoning", "url": "https://arxiv.org/pdf/2510.04022"}, {"title": "Position-aware Location Regression Network for Temporal Video Grounding", "url": "https://arxiv.org/pdf/2204.05499"}, {"title": "Enrich and Detect: Video Temporal Grounding with Multimodal LLMs", "url": "https://arxiv.org/pdf/2510.17023"}, {"title": "TEMPURA: Temporal Event Masked Prediction and Understanding for Reasoning in Action", "url": "https://arxiv.org/pdf/2505.01583"}, {"title": "Weakly-Supervised Multi-Level Attentional Reconstruction Network for Grounding Textual Queries in Videos", "url": "https://arxiv.org/pdf/2003.07048"}, {"title": "Video-XL-Pro: Reconstructive Token Compression for Extremely Long Video Understanding", "url": "https://arxiv.org/pdf/2503.18478"}]

Based on the search results, I found the specific performance metrics you mentioned for Qwen2.5-VL on the Charades-STA benchmark.

The Qwen2.5-VL-7B model achieves R@0.5 of 50.3%, R@0.7 of 24.3%, and mIoU of 43.6% on the Charades-STA temporal grounding benchmark. These metrics demonstrate the model's performance in localizing temporal moments within videos based on natural language queries without fine-tuning on the task (off-the-shelf performance).

The Charades-STA dataset is a standard benchmark for evaluating temporal video grounding capabilities, where models must identify the precise temporal boundaries of actions or events described in natural language within video clips. The metrics represent:

- **R@0.5**: Recall at 0.5 IoU (Intersection over Union) threshold
- **R@0.7**: Recall at 0.7 IoU threshold
- **mIoU**: Mean Intersection over Union

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-09T11:40:50.670Z)

Important finding. The RacketVision claim has a material flaw: the paper benchmarks ball tracking, racket pose, and trajectory forecasting ? NOT temporal action localization, rally segmentation, or highlight detection. The +19.2%/+14.6% mAP is ball-detection accuracy, not temporal event localization. The finding repeatedly cites RacketVision as validating "single sport-agnostic model" for the multisport temporal-localization claim, but it actually validates a different thing (multi-sport ball tracking). This would change the strength of the multisport-temporal claim.

The Qwen2.5-VL off-the-shelf numbers (R@0.5=50.3, R@0.7=24.3, mIoU=43.6) are confirmed.

Now let me verify the Gemma 4 claims (the 60s cap, no timestamp, Apache-2.0) and the conversational-QA / AVAS architecture claim. Let me also verify the ms-swift native video+audio GRPO claim since that's load-bearing for the framework recommendation.

13. user (2026-06-09T11:40:59.466Z)

Web search results for query: "Gemma 4 Google open model April 2026 Apache-2.0 video input length native audio"

Links: [{"title": "What Is Google Gemma 4? The Apache 2.0 Open-Weight Model With Native Audio and Vision | MindStudio", "url": "https://www.mindstudio.ai/blog/what-is-google-gemma-4-apache-open-weight"}, {"title": "Gemma 4: Byte for byte, the most capable open models", "url": "https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/"}, {"title": "Gemma 4 ? Google DeepMind", "url": "https://deepmind.google/models/gemma/gemma-4/"}, {"title": "Gemma (language model) - Wikipedia", "url": "https://en.wikipedia.org/wiki/Gemma_(language_model)"}, {"title": "What Is Gemma 4? Google's Apache 2.0 Open-Weight Model With Native Audio and Vision | MindStudio", "url": "https://www.mindstudio.ai/blog/what-is-gemma-4-google-apache-open-weight-model"}, {"title": "Gemma 4 12B: Specs, Benchmarks & How to Run It Locally", "url": "https://www.buildfastwithai.com/blogs/gemma-4-12b-guide"}, {"title": "Gemma 4 by Google: Specs, Benchmarks, Model Sizes, and How to Run It Locally (2026 Guide)", "url": "https://aurigait.com/blog/gemma-4-features-benchmarks-guide/"}, {"title": "Google's new open source Gemma 4 12B analyzes audio, video ? and runs entirely locally on a typical 16GB enterprise laptop - Technology data bank", "url": "https://www.dataworldbank.net/2026/06/03/googles-new-open-source-gemma-4-12b-analyzes-audio-video-and-runs-entirely-locally-on-a-typical-16gb-enterprise-laptop/"}, {"title": "Google's new open source Gemma 4 12B analyzes audio, video ? and runs entirely locally on a typical 16GB enterprise laptop | VentureBeat", "url": "https://venturebeat.com/technology/googles-new-open-source-gemma-4-12b-analyzes-audio-video-and-runs-entirely-locally-on-a-typical-16gb-enterprise-laptop"}, {"title": "Introducing Gemma 4 12B: a unified, encoder-free multimodal model", "url": "https://blog.google/innovation-and-ai/technology/developers-tools/introducing-gemma-4-12b/"}]

Based on the search results, here's what I found about Gemma 4:

Gemma 4 Overview

Gemma 4 was released in April 2025 under the Apache 2.0 license and is Google's most capable open-weight model family to date and the first to ship with native audio and vision capabilities alongside text.

Model Variants and Sizes

Gemma 4 comes in multiple sizes: 1B (designed for on-device use), 4B (the smallest multimodal variant), 12B (a mid-tier option with strong reasoning), and 27B (the flagship).

Native Audio Support

Gemma 4's audio support lets the model process spoken language directly from audio input – no separate speech-to-text step required, handling multilingual audio for transcription, translation, and audio-based Q&A applications. Gemma 4 supports native audio input on the E2B and E4B models.

Video Input

Short video clips can be passed as input to the 12B and 27B variants, enabling frame-level understanding and temporal reasoning.

Context Length

The larger Gemma 4 models offer up to 256K context window. The Gemma 4 12B has up to 256,000 tokens context with multilingual support, allowing it to process inputs like multi-minute video clips with synchronized audio.

Licensing

Gemma 4 is released under a commercially permissive Apache 2.0 license that provides complete developer flexibility and grants you complete control over your data, infrastructure, and models.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. user (2026-06-09T11:40:59.786Z)

AVAS Overview

AVAS is a VLM-powered video analytics system designed for long-form video understanding. Yes, it uses event knowledge graphs combined with retrieval-augmented generation and agentic reasoning.

Architecture

The system has two main components:

****Index Construction****: AVAS builds an Event Knowledge Graph (EKG) during a preprocessing phase. As the paper states: "the near real-time construction of Event Knowledge Graphs (EKGs) for efficient indexing of long or continuous video streams."

****Query Processing****: Rather than end-to-end VLM analysis, AVAS answers questions by retrieving from this index. The paper describes: "an agentic retrieval-generation mechanism that leverages EKGs to handle complex and diverse queries."

The architecture prioritizes durable indexing over direct VLM processing. The system uses "a

small VLM" to construct the EKG, then employs "agentic searching on graph" with "tri-view retrieval" (events, entities, visual embeddings) to find relevant information before generating answers.

This design enables AVAS to handle "videos spanning hundreds of hours" while maintaining near-real-time index construction speeds of "more than 5 FPS on typical edge servers."

15. assistant (2026-06-09T11:41:06.353Z)

AVAS is confirmed (event knowledge graph + RAG/agentic retrieval over a durable index, not end-to-end VLM QA) ? this supports the conversational-QA architecture recommendation.

The Gemma 4 claims need careful handling. The web search confirms Apache-2.0 and native audio, and confirms "short video clips" with the source noting "multi-minute video clips with synchronized audio" ? but there's a release-date discrepancy (the finding says April 2026, secondary sources say April 2025, and the finding's own caveat / project doc flags Gemma 4 as AMBER on post-cutoff secondary sources). The finding's "~60-second cap" specifically is what I should pin down. Let me check the official Gemma 4 model card, and verify the ms-swift native video+audio GRPO custom-reward claim (load-bearing for framework choice).

16. user (2026-06-09T11:41:11.488Z)

Gemma 4 Video Input Specifications

Maximum Video Duration

The model card specifies: **"Video supports a maximum of 60 seconds assuming the images are processed at one frame per second."**

Temporal Capabilities

The document does **not mention** temporal localization, timestamp output, or video moment/event grounding features. It only describes basic video understanding through frame-by-frame processing.

License

Apache 2.0 - The model card explicitly states: **"License: Apache 2.0"**

The document focuses on Gemma 4's ability to analyze video by processing sequences of frames, but provides no information about generating temporal annotations or identifying specific moments within videos.

17. user (2026-06-09T11:41:14.831Z)

Web search results for query: "ms-swift modelscope GRPO multimodal video custom reward function external plugin vLLM Qwen2.5-VL"

Links: [{"title": "Complete Multimodal GRPO Experiment Workflow ? swift 3.9.3 documentation", "url": "https://swift.readthedocs.io/en/v3.9/BestPractices/GRPO-Multi-Modal-Training.html"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 3.11.3 documentation", "url": "https://swift.readthedocs.io/en/v3.11/BestPractices/GRPO-Multi-Modal-Training.html"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 3.5.3 documentation", "url": "https://swift.readthedocs.io/en/v3.5/BestPractices/GRPO-Multi-Modal-Training.html"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 4.1.0.dev0 documentation", "url": "https://swift.readthedocs.io/en/latest/BestPractices/GRPO-Multi-Modal-Training.html"}, {"title": "ms-swift - Qwen", "url": "https://qwen.readthedocs.io/en/v2.5/training/RL/ms_swift.html"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 3.4.1.post1 documentation", "url": "https://swift.readthedocs.io/en/v3.4/BestPractices/GRPO-Multi-Modal-Training.html"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 3.7.3 documentation", "url": "https://swift.readthedocs.io/en/v3.7/BestPractices/GRPO-Multi-Modal-Training.html"}]

I found comprehensive documentation about this topic. Here's what I found:

MS-SWIFT supports multimodal GRPO training with Qwen2.5-VL-3B-Instruct, external plugins for custom reward functions, vLLM integration, and ModelScope datasets. The configuration shows how these components work together:

****Key Components:****

1. ****External Plugins****: The `--external_plugins` parameter references a plugin file (e.g., `examples/train/grpo/plugin/plugin.py`) that enables custom reward functions.
2. ****Reward Functions****: External reward functions like `external_r1v_acc` and `format` can be used as reward functions for training.
3. ****vLLM Integration****: The training configuration supports vLLM in server mode with specified host and port settings.
4. ****Model Selection****: Qwen2.5-VL-3B-Instruct is selected as the base model, with the Instruct version chosen over the base model to rapidly achieve format rewards.
5. ****Multi-GPU Support****: SWIFT GRPO training supports multi-GPU deployment to accelerate rollouts.

The search results show this is part of the official SWIFT documentation for multimodal GRPO experiment workflows, with examples demonstrating how to configure and run such training pipelines using ModelScope datasets.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

18. assistant (2026-06-09T11:41:22.663Z)

Key verifications complete:

- Gemma 4 official model card confirms the "60 seconds" video cap and "no temporal localization/timestamp" ? exactly as the finding claims. This disqualifies it as primary extractor (finding is correct).
- ms-swift GRPO multimodal with external_plugins custom reward + vLLM is confirmed in official docs. But I note the documented example uses Qwen2.5-VL-3B with `*image*` reward (`external_r1v_acc`). The finding claims it does this over `*video*` rollouts. Let me check whether the multimodal GRPO actually supports video inputs specifically, and verify the LLaMA-Factory/EasyR1 "image-only GRPO" claim since that's the load-bearing differentiator for the framework recommendation.

Let me verify the EasyR1 image-only claim and the ms-swift video-GRPO distinction, plus the InternVL3 license.

19. user (2026-06-09T11:41:29.439Z)

Video Support in Multimodal GRPO Workflow

Based on the provided documentation, **there is no mention of video inputs**. The workflow is entirely image-based.

The document does not contain any references to:

- Video inputs or processing
- Frames per second (fps) or frame counts
- Temporal grounding
- Video-specific parameters

All three tasks documented (ClevrCount, Geometric QA, and Multimodal Open R1) exclusively use image data. The dataset structures consistently reference "images" fields organized as lists of image paths: `["image_path1", "image_path2"]`. The implementation appears limited to static visual inputs rather than temporal video sequences.

20. user (2026-06-09T11:41:31.538Z)

Web search results for query: "EasyR1 hiyouga veRL video support GRPO multimodal image video RLHF framework"

Links: [{"title": "GitHub - hiyouga/EasyR1: EasyR1: An Efficient, Scalable, Multi-Modality RL Training Framework based on veRL", "url": "https://github.com/hiyouga/EasyR1"}, {"title": "GitHub - arvillion/EasyR1-video: EasyR1: An Efficient, Scalable, Multi-Modality RL Training Framework based on veRL", "url": "https://github.com/arvillion/EasyR1-video"}, {"title": "EasyR1/README.md at main · hiyouga/EasyR1", "url": "https://github.com/hiyouga/EasyR1/blob/main/README.md"}, {"title": "GitHub - Gabesarch/ViGoRL: EasyR1: An Efficient, Scalable, Multi-Modality RL Training Framework based on veRL", "url": "https://github.com/Gabesarch/ViGoRL"}, {"title": "LLaMA Factory on X: \"?Introducing EasyR1: an efficient, scalable, multi-modality RL training framework We have witnessed the remarkable success of GRPO algorithm in DeepSeek R1. Now we extended veRL @veRL_project to support vision-language models, enabling efficient RL training for Qwen2.5-VL https://t.co/9yrWD3igH4\" / X", "url": "https://x.com/llamafactory_ai/status/1893879214727991504"}, {"title": "EasyR1/assets/baselines.md at main · hiyouga/EasyR1", "url": "https://github.com/hiyouga/EasyR1/blob/main/assets/baselines.md"}, {"title": "Releases · hiyouga/EasyR1", "url": "https://github.com/hiyouga/EasyR1/releases"}, {"title": "EasyR1 - EasyR1 is an efficient, scalable RL training framework for multimodal models, \"?", "url": "https://jimmysong.io/ai/easy-r1/"}, {"title": "Workflow runs · hiyouga/EasyR1", "url": "https://github.com/hiyouga/EasyR1/actions"}, {"title": "Zooming-without-Zooming/EasyR1 at main · inclusionAI/Zooming-without-Zooming", "url": "https://github.com/inclusionAI/Zooming-without-Zooming/tree/main/EasyR1"}]

Based on the search results, here's what I found about EasyR1, hiyouga, veRL, video support, GRPO, and multimodal image/video RLHF framework:

Overview

EasyR1 is a clean fork of the original veRL project to support vision language models. EasyR1 is an efficient, scalable reinforcement learning (RL) training framework for multimodal models, built as a clean fork of veRL with engineering optimizations such as HybridEngine and vLLM SPMD support to enable RL training and evaluation for large language and vision-language models.

Key Features

EasyR1 is compatible with text and vision-text models and dataset formats. It supports GRPO, DAP0, Reinforce++, and optimizations like padding-free training.

GRPO Algorithm Support

The GRPO algorithm achieved remarkable success in DeepSeek R1, and was extended to veRL to support vision-language models.

Video Support

Regarding video support specifically, I found a repository fork called "EasyR1-video" on GitHub (arvillion/EasyR1-video), which appears to be an extension that adds video capabilities to the framework, though the search results don't provide extensive details about the specific video implementation features.

The framework appears to be actively developed by hiyouga and the community, supporting various multimodal reasoning tasks through RLHF (Reinforcement Learning from Human Feedback) techniques.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

21. assistant (2026-06-09T11:41:40.352Z)

Significant finding. This challenges the framework recommendation:

1. ****EasyR1 image-only claim is partly OUTDATED****: there's a community fork

`arvillion/EasyR1-video` adding video, and the main repo is actively developed. The finding's claim that EasyR1 is "image-only" is the load-bearing reason LLaMA-Factory drops to "medium." It may be becoming outdated, though the *main* hiyouga/EasyR1 README still describes "text and vision-text" (not explicitly video).

2. **ms-swift multimodal GRPO documented example is ALSO image-only**: The official "Complete Multimodal GRPO Experiment Workflow" page uses ClevrCount/Geometric QA/Multimodal Open R1 ? all images, no video, no fps/num_frames. This directly undercuts the finding's strongest differentiator that ms-swift is "the ONLY framework that runs multimodal GRPO with a custom external-reward plugin over VIDEO rollouts." The custom-reward-over-VIDEO-rollouts claim is not demonstrated in the doc the finding itself cites.

Let me verify whether ms-swift actually supports video in GRPO anywhere (the finding cites an issue #7326), and check the InternVL3 license and the Llama 4 / Qwen3-VL uniform-license claim. These are the remaining load-bearing items.

22. user (2026-06-09T11:41:47.461Z)

Analysis of GRPO and Video Training Discussion

Does this issue discuss video inputs with GRPO training?

Yes. The issue explicitly addresses "multi-frame visual input training" with "max_frames=180" using GRPO, stating: "I'm currently experimenting with **multi-frame visual input training** (max_frames=180) using **Megatron + GRPO**".

Does it confirm or deny GRPO works with video and custom rewards?

The issue neither confirms nor denies successful functionality. Instead, it reports a problem: the user encounters out-of-memory errors when attempting this combination. They note that "GPU memory runs out very quickly" and training remains stable only up to approximately 32 frames.

What is the resolution or status?

There is no resolution provided. The issue remains open without responses or marked solutions. The user poses questions seeking guidance on how to scale multi-frame training, asking: "how are **CP / SP / TP / PP** typically combined in practice to scale to high frame counts?" This indicates the issue documents an unresolved technical challenge rather than a confirmed working configuration.

23. user (2026-06-09T11:41:47.515Z)

License Information for InternVL3-8B

Primary License

The model itself is released under the **MIT License**.

Dependency License

The model uses Qwen2.5 as its language component, which operates under the **Apache-2.0 License**. As stated in the documentation: "This project uses the pre-trained Qwen2.5 as a component, which is licensed under the Apache-2.0 License."

License Dependency

Yes, the overall licensing situation depends on the LLM backbone. Users must comply with both the MIT license for InternVL3-8B and the Apache-2.0 license for the Qwen2.5 component.

Video Timestamp Output

The model card does **not** mention video temporal grounding or timestamp output capabilities. While the documentation describes video understanding functionality and provides code examples for video processing, there is no reference to generating temporal annotations or timestamps for video content.

24. assistant (2026-06-09T11:41:57.984Z)

More nuance:

- ms-swift issue #7326 confirms users ARE running multi-frame/video GRPO (max_frames=180) with Megatron+GRPO ? so ms-swift video-GRPO is *attempted/possible*, but the issue reports OOM beyond ~32 frames and is UNRESOLVED. So the capability exists but the "runs over video rollouts" claim is real-but-fragile (VRAM-bound), which actually aligns with the finding's own VRAM caveat. The finding's core differentiator (ms-swift supports video GRPO) is supported in direction, though the cleanest *official doc* example is image-only.
- InternVL3-8B is MIT + Apache-2.0 (Qwen2.5 backbone), no native timestamp grounding ? exactly as the finding claims. Confirmed.

Now the remaining load-bearing items: (1) Qwen3-VL uniform Apache-2.0 across ALL sizes including the 235B (the finding's "no per-size license fork" claim, which it flagged as not individually verified). (2) Llama 4 MAU-cap license claim. Let me verify the 235B and 2B/4B cards, and the Llama license. Let me also sanity-check the InternVL3.5 temporal-grounding claim since it bears on the InternVL fallback.

25. user (2026-06-09T11:42:01.739Z)

Based on the web page content provided, the license field for Qwen3-VL-235B-A22B-Instruct explicitly states:

```
***"License: apache-2.0"***
```

This is a standard Apache License 2.0 with no additional MAU (Monthly Active Users) caps or custom restrictions mentioned on the page.

26. user (2026-06-09T11:42:03.014Z)

Based on the webpage content provided, the license field for the Qwen3-VL-2B-Instruct model lists "apache-2.0".

The relevant section states: "License: apache-2.0"

27. user (2026-06-09T11:42:09.021Z)

Web search results for query: "Llama 4 license community license MAU 700 million monthly active users acceptable use commercial restriction"

Links: [{"title": "Llama AI for Commercial Use: License & Enterprise Impact", "url": "https://llamaaimodel.com/commercial-use/"}, {"title": "Significant Risks in Using AI Models Governed by the Llama License ? Open Source Guy", "url": "https://shujisado.org/2025/01/27/significant-risks-in-using-ai-models-governed-by-the-llama-license/"}, {"title": "Llama 4 Community License Agreement", "url": "https://www.llama.com/llama4/license/"}, {"title": "Llama 4 Guide: Scout, Maverick, Behemoth Status & Muse Spark (2026)", "url": "https://codersera.com/blog/llama-4-complete-guide-2026/"}, {"title": "Llama Commercial License Guide 2025 | Business Use Terms", "url": "https://artificial-intelligence-wiki.com/generative-ai/large-language-models/llama-commercial-license/"}, {"title": "Rohan Paul on X: \"Llama 4 License: A custom commercial license, the Llama 4 Community License Agreement, is available at: github.com/meta-llama/llama-models/blob/main/models/llama4/LICENSE The key takeaway is that while you get broad, free rights to use, modify, and distribute Llama 4, there's https://t.co/NzIPryP6U\" / X", "url": "https://x.com/rohanpaul_ai/status/1908630912385892731"}, {"title": "Llama FAQs", "url": "https://www.llama.com/faq/"}, {"title": "Meta Llama 4 explained: Everything you need to know", "url": "https://www.techtarget.com/whatis/feature/Meta-Llama-4-explained-Everything-you-need-to-know"}, {"title": "Meta's new A.I. is an open-source breakthrough with fine print to freeze out competitors", "url": "https://finance.yahoo.com/news/meta-open-source-breakthrough-fine-025211942.html"}, {"title": "llama4/license", "url": "https://ollama.com/library/llama4/blobs/399a8a5a36db"}]

Based on the search results, here's what I found about the Llama 4 license regarding the 700 million MAU threshold and commercial restrictions:

Key License Terms

The Llama 4 Community License Agreement is not a traditional open-source license; it is a sophisticated legal instrument designed to build a commercial ecosystem on Meta's terms. The license is permissive, granting your business a royalty-free, worldwide right to use the Llama models in your commercial products and services.

The 700 Million MAU Restriction

If your service exceeds 700M MAUs, your right to use Llama 4 under the standard community license terminates. You would need to proactively contact Meta to request a special, separate license. Meta may grant or deny this request at its "sole discretion," and would likely negotiate custom commercial terms.

This provision is designed to prevent hyperscale competitors from leveraging Meta's R&D without additional agreements.

Definition of MAU

MAU means "monthly active users" that access or use your (and your affiliates') products and services. Examples include users accessing an internet-based service and monthly users/customers of the licensee's software applications or hardware devices.

Acceptable Use Policy

Your use of the Llama Materials must comply with applicable laws and regulations (including trade compliance laws and regulations) and adhere to the Acceptable Use Policy for the Llama Materials.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

28. assistant (2026-06-09T11:42:15.274Z)

All confirmed:

- Qwen3-VL is apache-2.0 uniformly across 2B, 8B, and 235B (no per-size fork, no MAU cap) ? the finding's "uniform Apache-2.0" claim is now verified on the smallest and largest sizes.
- Llama 4 Community License has the 700M MAU cap and acceptable-use restriction ? exactly as the finding claims. Confirmed it's NOT clean.

I have enough to assess all the load-bearing claims across the three focus areas. Let me do one final verification on the most decision-relevant remaining claim: the finding's central skeptical thesis that "VLMs capture coarse temporal regions, not accurate boundaries" ? confirm this is the actual literature consensus and that the best fine-tuned VLM genuinely tops out near R@0.7?50%. The Time-R1 number is confirmed; let me confirm the survey-level consensus quote.

29. user (2026-06-09T11:42:28.189Z)

VTG-MLLMs and Fine-Grained Temporal Boundary Precision

The survey does acknowledge limitations of MLLMs for precise temporal grounding. The key passage states:

> "standard MLLMs often struggle to capture fine-grained temporal dependencies. This challenge largely stems from their vision encoders treating video as an unordered 'bag-of-features,' which discards the crucial sequential information essential for precise event localization."

This appears in Section 3.1.2 (Executors), explaining why MLLMs require architectural enhancements and training optimization to perform temporal boundary prediction effectively.

However, the survey **does not explicitly compare MLLMs against traditional temporal action

localization (TAL) methods** to determine if specialized TAL approaches remain more competitive for boundary precision. The document focuses on VTG-MLLM approaches but does not provide head-to-head evaluations against conventional, non-MLLM-based TAL methods to establish whether traditional approaches offer superior boundary precision.

The survey emphasizes that architectural innovations and sophisticated training strategies are necessary to address the fundamental limitations MLLMs face with temporal reasoning, but it doesn't position traditional TAL methods as superior alternatives.

30. assistant (2026-06-09T11:42:43.392Z)

The survey confirms the core skeptical claim: MLLMs "struggle to capture fine-grained temporal dependencies... discards the crucial sequential information essential for precise event localization." However, one nuance: the survey does NOT itself do a head-to-head MLLM-vs-traditional-TAL comparison. The finding's claim that "specialized TAL heads remain ahead on boundary precision" rests on comparing different benchmarks (ActionFormer ~66.8% mAP on THUMOS14 vs VLM R@0.7=50.1% on Charades-STA) ? which are different datasets, so it's a cross-benchmark inference, not a direct comparison. The finding does flag this in its own caveats ("Charades-STA/THUMOS14/ActivityNet are open-domain... absolute numbers won't map 1:1").

I have thoroughly verified the load-bearing claims across all three focus areas. Let me compile the structured output.

Summary of my verification across the three focus areas:

****1. VLM-can-reach-high-P/R-temporal-localization claim**** ? The finding's *skeptical* framing is well-supported: best fine-tuned VLM (Time-R1) tops out at R@0.7=50.1% (confirmed), Qwen2.5-VL off-the-shelf R@0.7=24.3% (confirmed), and the survey confirms MLLMs "struggle to capture fine-grained temporal dependencies." Qwen3-VL's Apache-2.0 license + Text-Timestamp Alignment + 256K?1M context are all confirmed on primary HF cards. The finding correctly does NOT claim a VLM reaches high P/R ? it argues the opposite, and that is supported.

****2. Single-model-multisport claim**** ? PARTIALLY REFUTED as cited. RacketVision (AAAI'26 Oral, confirmed) shows +19.2%/+14.6% mAP from unified multi-sport training, but this is ****ball-tracking detection mAP, NOT temporal localization / rally segmentation****. The finding repeatedly uses RacketVision to validate a single sport-agnostic ***temporal-localization*** model, which the paper does not measure. The single-model concept is plausible, but the cited evidence supports multi-sport ***ball tracking***, not multi-sport ***rally/highlight temporal extraction***.

****3. Conversational-QA-via-event-store**** ? Supported. AVAS (event knowledge graph + RAG/agentic retrieval over a durable index) is confirmed as a real system matching the recommended architecture.

31. user (2026-06-09T11:43:47.033Z)

Output does not match required schema: root: must have required property 'checked'

32. assistant (2026-06-09T11:43:50.089Z)

The error suggests the StructuredOutput parameters weren't passed correctly. Let me retry with the exact schema shape.

33. user (2026-06-09T11:44:39.253Z)

Output does not match required schema: root: must have required property 'checked', root: must have required property 'corrections'

34. assistant (2026-06-09T11:44:42.871Z)

The schema validator reports `checked` and `corrections` as missing even though I passed them. This is likely a payload-size truncation issue ? the large array values may be getting cut off in transit. Let me shorten the field contents substantially while keeping the load-bearing verdicts and corrections intact.

35. user (2026-06-09T11:45:18.173Z)

Output does not match required schema: root: must have required property 'checked', root: must have required property 'corrections'

36. assistant (2026-06-09T11:45:22.351Z)

The validator still reports `checked` and `corrections` missing. The structure looks correct, so the issue may be a stray trailing `` artifact I accidentally included, or a parser problem with the long strings. Let me retry cleanly with materially shorter values to rule out a size/encoding issue.

37. user (2026-06-09T11:45:48.192Z)

Output does not match required schema: root: must have required property 'checked', root: must have required property 'corrections'

38. assistant (2026-06-09T11:45:53.627Z)

The validator keeps reporting both required array properties as missing despite being present, and I notice my last two calls had a stray `` / extra `` artifact in the trailing text. The harness may be mis-parsing the call. Let me try a minimal version to isolate whether the issue is structural ? using just the `focus` plus one short entry each in `checked` and `corrections`.

39. user (2026-06-09T11:45:56.210Z)

Output does not match required schema: root: must have required property 'checked', root: must have required property 'corrections'

40. assistant (2026-06-09T11:46:00.608Z)

Even the minimal version fails identically, which confirms the problem is not content size ? the `checked` and `corrections` arguments aren't reaching the validator at all. I suspect the trailing `` and `` artifacts in my parameter block are corrupting the parse. Let me write the call with absolutely clean parameter blocks and no stray closing tags inside any value.

41. user (2026-06-09T11:46:01.262Z)

Structured output provided successfully

42. assistant (2026-06-09T11:46:05.105Z)

The minimal call succeeded ? confirming the earlier failures were caused by malformed parameter blocks (the stray `` / `` artifacts I accidentally embedded). Now let me submit the full, correct verification with clean parameter blocks.

43. user (2026-06-09T11:46:47.629Z)

Output does not match required schema: root: must have required property 'checked', root: must have required property 'corrections'

44. assistant (2026-06-09T11:46:51.586Z)

The minimal call succeeded but the full one fails again ? so it IS a size/length threshold issue with the large `checked` / `corrections` payloads getting truncated before validation. I need to materially shorten each entry's strings to stay under the limit while preserving the load-bearing verdicts. Let me trim aggressively.

45. user (2026-06-09T11:47:13.807Z)

Output does not match required schema: root: must have required property 'checked', root: must

have required property 'corrections'

46. assistant (2026-06-09T11:47:18.084Z)

I again included a stray ``</invoke>`` / ``</parameter>`` artifact at the end of the ``corrections`` block in that call ? that trailing junk is corrupting the tool call so the two array params never register. The minimal call that worked had no such artifact. Let me submit with clean blocks and no trailing tags inside the value.

47. user (2026-06-09T11:47:39.673Z)

Output does not match required schema: root: must have required property 'checked', root: must have required property 'corrections'

48. assistant (2026-06-09T11:47:47.375Z)

I keep accidentally appending a stray ``</invoke></parameter>`` after the ``corrections`` value, which breaks the call. The successful minimal call did not have that. I must end the ``corrections`` parameter block cleanly. Let me submit once more, being careful that the final parameter closes with exactly ``</parameter>`` and then the single ``</invoke>`` with nothing extra.

49. user (2026-06-09T11:48:09.903Z)

Output does not match required schema: root: must have required property 'checked', root: must have required property 'corrections'

50. assistant (2026-06-09T11:48:15.352Z)

Still failing with clean blocks this time, which means it is genuinely a payload-length cutoff truncating the two large arrays. The minimal call worked. I'll cut the content roughly in half ? fewer words per field, drop the two lowest-value entries (Llama 4 and the LOVO-provenance, both clearly "supported" and not focus-critical) to get under the threshold, keeping all three focus clusters and the corrections.

51. user (2026-06-09T11:48:33.320Z)

Output does not match required schema: root: must have required property 'checked', root: must have required property 'corrections'

52. assistant (2026-06-09T11:48:37.988Z)

Still failing even when shortened. The minimal call worked, so let me bisect: I'll add back to the working minimal call ONE field at a time. First, test whether a full-size ``checked`` array (10 entries) alone works with a trivial ``corrections``.

53. user (2026-06-09T11:48:40.058Z)

Structured output provided successfully

54. assistant (2026-06-09T11:48:51.628Z)

A 5-entry ``checked`` with short strings works. So the failure is a total-payload size threshold. The real content is too long. I need to compress hard. Let me bisect the size boundary: test the full real content but trimmed to ~5 entries and short corrections to find what fits, then prioritize the focus-critical entries (RacketVision refuted, ms-swift uncertain, EasyR1 outdated, plus the VLM-ceiling and AVAS supports).

Let me try the real content but capped at the 6 most decision-relevant entries with tight strings.

55. user (2026-06-09T11:49:10.524Z)

Output does not match required schema: root: must have required property 'checked', root: must

have required property 'corrections'

56. assistant (2026-06-09T11:49:14.890Z)

Still failing at 6 condensed entries + 4 corrections. The 5-trivial-entry test passed. The threshold is between those. Let me find it: test the real 6-entry content but with the corrections trimmed to a single short string, to confirm `checked` size is the dominant factor.

57. user (2026-06-09T11:49:26.494Z)

Output does not match required schema: root: must have required property 'checked', root: must have required property 'corrections'

58. assistant (2026-06-09T11:49:35.299Z)

Even with a single short correction, the 6 real entries fail. So the `checked` array content is over the size limit. The 5-trivial-entry version passed. Let me test the exact boundary: submit the 5 most decision-critical real entries (drop one) with a single merged correction.

59. user (2026-06-09T11:49:48.084Z)

Output does not match required schema: root: must have required property 'checked'

60. assistant (2026-06-09T11:49:51.589Z)

Progress ? this time only `checked` is flagged (not `corrections`), meaning `corrections` now fit and `checked` is just over the line. I need to trim `checked` a bit more. Let me shorten the evidence/claim strings in the 5 entries further.

61. user (2026-06-09T11:50:03.638Z)

Output does not match required schema: root: must have required property 'checked'

62. assistant (2026-06-09T11:50:07.493Z)

Still just over on `checked`. Let me trim one more notch ? compress the longest two entries' evidence strings.

63. user (2026-06-09T11:50:18.424Z)

Output does not match required schema: root: must have required property 'checked'

64. assistant (2026-06-09T11:50:23.769Z)

Still over. The cumulative size across focus+checked+corrections matters. Let me drop the `focus` to a short sentence and the corrections to fewer/shorter, freeing budget for `checked`. I'll also drop to compact entries.

65. user (2026-06-09T11:50:33.222Z)

Output does not match required schema: root: must have required property 'checked', root: must have required property 'corrections'

66. assistant (2026-06-09T11:50:37.631Z)

I dropped `corrections` entirely there (it's required), and `checked` is still flagged. Let me keep both required fields but make `checked` shorter still and `corrections` minimal. The 5-trivial-entry test passed at small size; my real entries are roughly 2-3x longer each. Let me reduce to compact single-line entries.

67. user (2026-06-09T11:50:45.567Z)

Output does not match required schema: root: must have required property 'checked', root: must

have required property 'corrections'

68. assistant (2026-06-09T11:50:48.882Z)

I keep omitting `corrections` ? it's required and I dropped it again. Both `checked` and `corrections` must be present. Let me include a minimal `corrections` and keep `checked` compact.

69. user (2026-06-09T11:50:55.902Z)

Output does not match required schema: root: must have required property 'checked', root: must have required property 'corrections'

70. assistant (2026-06-09T11:50:59.712Z)

I forgot `corrections` AGAIN ? I'm pasting only two parameter blocks. Both `checked` and `corrections` are required and I must include the `corrections` parameter block. Let me add it.

71. user (2026-06-09T11:51:07.122Z)

Output does not match required schema: root: must have required property 'checked'

72. assistant (2026-06-09T11:51:11.991Z)

Only `checked` now flagged ? `corrections` fit. `checked` is still marginally over. Let me shave the `checked` evidence strings a little more.

73. user (2026-06-09T11:51:19.253Z)

Output does not match required schema: root: must have required property 'checked'

74. assistant (2026-06-09T11:51:25.582Z)

Still over by a small margin. Let me drop to 4 entries by merging the VLM-ceiling and Qwen3-VL entries, keeping the focus-critical refuted/uncertain verdicts intact.

75. user (2026-06-09T11:51:33.886Z)

Output does not match required schema: root: must have required property 'checked'

76. assistant (2026-06-09T11:51:41.047Z)

Still over with 4 entries. The total request size seems to be the constraint and my entries are still too verbose. Let me cut the corrections to one short line and make checked entries very terse.

77. user (2026-06-09T11:51:47.549Z)

Output does not match required schema: root: must have required property 'checked'

78. assistant (2026-06-09T11:51:53.832Z)

Still over. This is stubborn ? there may be a hard per-string or per-field byte cap rather than total size. Given the 5-trivial-entry test passed (very short strings), the cap is likely on cumulative `checked` size and my strings remain too long. Let me go maximally terse: 4 ultra-short entries.

79. user (2026-06-09T11:51:58.073Z)

Output does not match required schema: root: must have required property 'checked'

80. assistant (2026-06-09T11:52:03.889Z)

Even ultra-terse 4 entries fail, while the bisect-test A (5 entries, `c1`..`s5`) passed. The difference: test A had `corrections: ["c1"]` (tiny). My current `corrections` is a long single string. Let me confirm by shrinking `corrections` to a trivial value and keeping the terse `checked`.

81. user (2026-06-09T11:52:06.358Z)

Structured output provided successfully

82. assistant (2026-06-09T11:52:10.810Z)

That worked. So the `corrections` long string was pushing it over. The constraint is total payload. Now I know the budget: terse 4-entry `checked` + tiny `corrections` fits. Let me rebalance ? keep the terse `checked` and use a slightly larger but still-bounded `corrections` (the focus-critical caveats). Let me incrementally grow corrections to fit.

83. user (2026-06-09T11:52:19.224Z)

Output does not match required schema: root: must have required property 'checked'

84. assistant (2026-06-09T11:52:25.516Z)

Over again ? the longer `checked` strings pushed it over. The working version had the very terse 4-entry `checked`. Let me keep that exact terse `checked` and only expand `corrections` to the three focus-critical caveats, which is where the load-bearing recommendation-changing content lives.

85. user (2026-06-09T11:52:30.514Z)

Output does not match required schema: root: must have required property 'checked'

86. assistant (2026-06-09T11:52:37.079Z)

The terse `checked` + this `corrections` is over, but terse `checked` + `["x"]` worked. So `corrections` here is too big. Let me trim `corrections` to two shorter strings and keep the terse `checked`.

87. user (2026-06-09T11:52:41.017Z)

Output does not match required schema: root: must have required property 'checked'

88. assistant (2026-06-09T11:52:43.649Z)

Still over. The boundary is tight. The `["x"]` version worked. Let me trim corrections to one moderate string.

89. user (2026-06-09T11:52:47.641Z)

Output does not match required schema: root: must have required property 'checked'

90. assistant (2026-06-09T11:52:51.557Z)

The `focus` field is also part of the budget and mine is longer than test C's "test C". Let me shrink `focus` drastically and keep this corrections.

91. user (2026-06-09T11:52:55.271Z)

Output does not match required schema: root: must have required property 'checked'

92. assistant (2026-06-09T11:52:59.282Z)

Shrinking `focus` didn't help, which means the binding constraint is the `checked`+`corrections` combined size, not `focus`. The version that worked had `corrections:["x"]` (3 chars). My

corrections string is ~180 chars. The gap is small. Let me cut corrections to ~60 chars.

93. user (2026-06-09T11:53:02.264Z)

Output does not match required schema: root: must have required property 'checked'

94. assistant (2026-06-09T11:53:08.585Z)

Still failing even with ~70-char corrections and terse checked. But test C with the IDENTICAL terse `checked` and `corrections:["x"]` worked. The only difference now is the corrections string length (~70 vs 1 char). That ~70 chars should be trivial. This inconsistency suggests the failure may not be pure size ? it may be that the validator intermittently truncates, OR a specific character is breaking parsing. My corrections contain `=` and `-` and a hyphen-heavy phrase. Test C corrections was `["x"]`. Let me test whether the terse checked + a plain-word corrections (no special chars) works.

95. user (2026-06-09T11:53:12.202Z)

Output does not match required schema: root: must have required property 'checked'

96. assistant (2026-06-09T11:53:18.739Z)

Removing special chars didn't help. So it's genuinely a cumulative size threshold, and my 4 real entries (even terse) plus a 2-line corrections exceed it, while 4 entries + `["x"]` fit. The budget for `corrections` beyond the 4-entry `checked` is very small. Let me find the max corrections that fits: keep terse `checked`, corrections = one ~40-char string.

97. user (2026-06-09T11:53:21.857Z)

Structured output provided successfully

98. assistant (2026-06-09T11:53:38.341Z)

That succeeded. So the budget is real and tight: 4 terse entries + one ~42-char correction fits. The earlier 2-line corrections tipped it over.

Now I have a successfully-recorded structured output, but it's heavily compressed and lost the ms-swift and Gemma corrections plus the rich evidence. The instructions say call StructuredOutput once; I've now succeeded. However the recorded version is minimal. Since the script reads the structured output, and I've now landed a valid one, I should ensure the most decision-critical content is captured. The successful call captured: the 4 verdicts (including the refuted RacketVision and uncertain ms-swift) and the top correction. That preserves the load-bearing findings: the RacketVision overclaim (refuted) and ms-swift overstatement (uncertain), plus the supported VLM-ceiling, Qwen3-VL/Gemma4 facts.

The structured output has been recorded successfully. My adversarial verification is complete.

Summary of what was verified against primary sources and captured in the structured output:

****Focus area 1 ? VLM high-P/R temporal localization:**** SUPPORTED as the findings frame it (skeptically). Best fine-tuned VLM (Time-R1) tops out at R@0.7?50.1 on Charades-STA; off-the-shelf Qwen2.5-VL-7B is R@0.7=24.3; the survey confirms MLLMs struggle with fine-grained boundaries. The findings correctly do NOT claim a VLM reaches high P/R. Qwen3-VL's Apache-2.0 license (verified on 2B/8B/235B cards), Text-Timestamp Alignment, and 256K?1M context are all confirmed. Gemma 4's 60s-cap/no-timestamp disqualifier is confirmed on the official model card.

****Focus area 2 ? single-model multisport:**** The key correction. RacketVision (AAAI'26), cited as the strongest evidence for a single multisport temporal-localization model, only benchmarks ball tracking, racket pose, and trajectory forecasting ? NOT temporal action localization. The +19.2%/+14.6% mAP is ball-detection accuracy. Verdict: REFUTED as cited. The single-model recommendation can still stand on architectural grounds, but multisport temporal generalization is asserted-not-measured.

****Focus area 3 ? conversational-QA-via-event-store:**** SUPPORTED. AVAS confirms the event-

knowledge-graph index + agentic retrieval architecture (extract-once-index-then-retrieve), matching the recommendation.

****Secondary:**** ms-swift's "ONLY framework with video custom-reward GRPO" is overstated (official GRPO doc is image-only; video-GRPO only in an unresolved OOM issue) and the EasyR1-image-only basis for downgrading LLaMA-Factory is going stale (a video fork exists) ? verdict UNCERTAIN/OUTDATED.

Note: I had to heavily compress the structured payload because the tool rejected larger outputs at a size threshold; the recorded output preserves the load-bearing verdicts (RacketVision refuted, ms-swift uncertain) and the primary correction. The fuller evidence and the additional corrections (Gemma date misattribution; TAL-vs-VLM being a cross-benchmark inference not a head-to-head) are documented in this message for the record.